Retained Material, Work Clocks and Renewal Certificates

Technical Supplement to Q-RAIF C

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Technical documentation for Q-RAIF C main paper (`QRAIF_C_manuscript.pdf`). This supplement contains the paper's complete technical appendices, not a new paper or an additional scientific result. The assumptions, open joins and scope of each certificate are retained. Appendix letters and all pre-existing result, equation, figure, table and citation numbers agree with the main paper's complete source. References to main sections and results link to the companion PDF. The main paper's certificate guide maps these proofs to its claims.

A From a charged material to a certified finite interface

A material action can retain the charged fields rather than replace them by neutral oscillators. The operator boson–fermion correspondence gives a convenient construction on a circle of circumference L. For one Dirac flavor with antiperiodic spatial fermions, its current oscillators and integer chiral charges have free energy

$$H_0 = \frac{\pi}{L}(N_R^2 + N_L^2) + \sum_{k\geq 1} \frac{2\pi k}{L}(m_{R,k} + m_{L,k}).$$

The current descendants of the charge vacua form a complete orthonormal basis of the fermion Hilbert space [7, Sections 2.5–2.6]. The correspondence includes Klein factors and charged odd operators; it is stronger than equality of partition functions.

For a scalar ϕ of period $\sqrt{\pi}$, put $w = N_R + N_L$ and $n = N_R - N_L$. Its zero-mode description is

$$\bigoplus_{w\in\mathbb{Z}} L^2\big([0,\sqrt{\pi}],\, \psi(\phi_0 + \sqrt{\pi}) = (-1)^w \psi(\phi_0)\big) \otimes \mathcal{F}_{\mathrm{osc}}.$$

Here $p_0 = \sqrt{\pi} n$, $n \equiv w \pmod 2$, and the charge energy is $\pi(w^2 + n^2)/(2L)$. The state $w = 1, n = 0$ would have $N_R = N_L = 1/2$ and is excluded. An ordinary compact scalar with periodic zero-mode wavefunctions has different sectors. Fixed-spin bosonization requires this additional spin information [8, Section 3.5].

Take two such flavors and real material fields Z, θ, y. The local classical coframe is $d\phi_a + \lambda_a \theta f(Z) dZ$, together with $m_Z dZ, \sqrt{I}\, d\theta, \sqrt{J}\, dy$, where $m_Z^2, I, J > 0$. Its positive canonical energy is

$$\begin{aligned} H_{\mathrm{mat}} = {} & \frac{1}{2}\int \left[\sum_a \Pi_a^2 + \frac{(\Pi_Z - \sum_a \lambda_a \theta f(Z)\Pi_a)^2}{m_Z^2} + \frac{\Pi_\theta^2}{I} + \frac{\Pi_y^2}{J}\right] dx \\ & + \frac{1}{2}\int \left[\sum_a (\phi_a' + \lambda_a \theta f(Z) Z')^2 + m_Z^2 Z'^2 + I\theta'^2 + Jy'^2 + 2V\right] dx. \end{aligned}$$

Assume smooth f and a smooth nonnegative V independent of the scalar zero modes. A Fourier regulator retains modes $0 < |k| \leq K$ of the interaction and material fields, all winding and charge sectors, and the complete free oscillator tail above K. The real Fourier coordinates are not separately wrapped angles.

Proposition 22 (Positive charged material at finite resolution)**.** *For each finite K, the displayed ordered sum of squares has a nonnegative self-adjoint Friedrichs realization. With the free tail retained, the known boson–fermion unitary transports it to the full four-chiral-field CAR Hilbert space tensor the material space. It strongly commutes with all four canonical chiral charges. No small-coupling bound on the finite λ_a is needed.*

Proof. Work first in each winding sector on smooth zero-mode line-bundle sections and compactly supported smooth wavefunctions of the finite real coordinates. The kinetic factor

$$D_Z(x) = \Pi_Z(x) - \sum_a \lambda_a \theta(x) f(Z(x)) \Pi_a(x)$$

is symmetric: its extra coefficient is independent of every ϕ_a coordinate, so its formal divergence vanishes. The other momentum factors are symmetric and the spatial factors are real multiplication operators. Their integrated squares and V define a densely defined nonnegative symmetric operator on the common core. Its Friedrichs extension exists. Finite sums of winding sectors give a dense core for their direct sum. Add the positive free-tail Hamiltonian and use the complete current/charge basis to transport this operator to the fermion space. The smeared fermion operators themselves are transported, not replaced by cutoff vertex exponentials, so their CAR are retained. Translations of each $\phi_{a,0}$ leave the core and closed form invariant, as do winding phases. The associated charge unitaries therefore commute with the self-adjoint realization. □

Low-mode canonical squares retain their common zero-point constant; the free tail uses one fixed normal ordering. No coupling-dependent vacuum subtraction is made. The material momenta, including the shift of Π_Z, remain in the operator. Fourier resolution acts on the coupling, not on the carrier's charges or occupation numbers. It is spatially nonlocal at finite K. Removing this cutoff while controlling propagation, renormalized stress and interacting thermal preparations is a separate limit.

The local current also fixes which coupling is intended. With signature $(-,+)$ and $\epsilon^{01}=1$, let $j^\mu=\epsilon^{\mu\nu}\partial_\nu\phi/\sqrt{\pi}$ and $a_\mu=\lambda\theta f(Z)\partial_\mu Z$. Then

$$-a_\mu\partial^\mu\phi=A_\mu j^\mu,\qquad A_\mu=\sqrt{\pi}\,\epsilon_{\mu\nu}a^\nu.$$

For a static texture, $A_0=-\sqrt{\pi}\lambda\theta f(Z)Z'$ and $A_1=0$, giving the required density potential. The coframe's $-a_\mu a^\mu/2$ term and momentum shifts accompany it. The local dictionary does not by itself fix global holonomies or spin weights [9, Eq. (9) and Section III]; the construction above uses the specified operator correspondence.

Rotating (ϕ_1,ϕ_2) to active and spectator Gaussian modes must also rotate their charge lattice. For instance $(w_1,w_2)=(1,0)$ becomes $(1/\sqrt{2},1/\sqrt{2})$, not two independent integer windings. This scalar normal-mode change is different from rotating the actual fermion flavors within each direction. A Gaussian diagonalization cannot discard their spin constraint or their charged operators.

This material also admits finite autonomous record-controlled operation. Extend its Hamiltonian H_K trivially over finite degenerate record registers. For a bounded self-adjoint permitted operator C and real $h\in C_c^\infty$, set

$$A=\int h(s)e^{isH_K}Ce^{-isH_K}ds,\qquad V=e^{i\ell A}.$$

Integration by parts gives $\|[H_K,A]\|\le\|h'\|_1\|C\|$ and $\|[H_K,[H_K,A]]\|\le\|h''\|_1\|C\|$. The energy-change estimates preceding Proposition 53 now apply with the *interacting material* H_K, so finite products meet Proposition 50. Record-controlled C are finite block sums of the permitted operators; all previous reference systems remain in its input purification.

For $0\le g\in C^\infty(S^1)$, weighting the regulated coupled-sector energy density above by g gives a positive form Q_g with $0\le Q_g\le\|g\|_\infty H_K$. It is a closable sum of the same weighted differential and multiplication squares. Its actual expectation during autonomous execution obeys Eq. (117), with $a=\|g\|_\infty$ and $d=0$. This supplies a state and energy certificate on

the charged material itself, without a hard physical energy cutoff. It concerns the regulated energy test: a renormalized covariant stress, spatially local history clock and interacting thermal preparation need their own estimates. Gate permission and the finite-rms initial state are specified inputs, not products of this smoothing argument.

A.1 A finite material mode and its remainder

The finite mechanical coordinates need not arise from fields uniform over the whole line. A local binding potential supplies normalizable angle and storage modes, with the discarded modes still present. Here we specify a fixed-texture Gaussian realization; its state, source and thermal estimates are not an equivalence theorem for the preceding nonlinear charged regulator. In normalized fields $u = (\sqrt{I}\theta, \sqrt{J}y)$, let

$$A_{\mathrm{m}} = \begin{pmatrix} k/I & -k/\sqrt{IJ} \\ -k/\sqrt{IJ} & (k+K)/J \end{pmatrix}, \qquad \alpha = \lambda_{\min}(A_{\mathrm{m}}) > 0.$$

Choose $0 < s < 1$ and $\mu > 0$ with $\alpha > s\mu^2$. The material operator

$$L_{\mathrm{m}} = -\partial_x^2 + A_{\mathrm{m}} + s^2\mu^2 I - s(s+1)\mu^2 \operatorname{sech}^2(\mu x) I$$

has a strictly positive pointwise mass matrix. For a supplied kink $Z_0 = z_0 \tanh(\mu x)$, the well is $1 - Z_0^2/z_0^2$. Using that polynomial in the material potential keeps it nonnegative for all real Z, since $1 - Z^2/z_0^2 \le 1$. The well can therefore belong to the same positive local action; the kink's preparation and position fluctuations remain physical inputs.

Proposition 23 (Finite modes and their reaction bound)**.** *The normalized $b(x) = N \operatorname{sech}^s(\mu x)$ gives two material modes with squared-frequency matrix A_{m}. The orthogonal material space has gap $m_{\mathrm{gap}}^2 = \alpha + s^2\mu^2$. For compact real $a(x)$, the complete Gaussian form is*

$$q[\phi, u] = \|\phi' + a u_\Theta\|^2 + \langle u, L_{\mathrm{m}} u\rangle.$$

Retaining the carrier and $u = b q_{\mathrm{m}}$ gives

$$q_P[\phi, q_{\mathrm{m}}] = \|\phi' + ab q_\Theta\|^2 + q_{\mathrm{m}}^T A_{\mathrm{m}} q_{\mathrm{m}}.$$

For $z < m_{\mathrm{gap}}^2$, elimination of the orthogonal material space has self-energy $\Sigma(z) \ge 0$, with

$$\langle \chi, \Sigma(z)\chi\rangle \le \frac{\|Q(a\phi' + a^2 b q_\Theta, 0)\|^2}{m_{\mathrm{gap}}^2 - z}.$$

Here Q projects off b in both material components, and the exact retained operator is $K_{PP} - \Sigma(z)$.

Proof. For $D_s = \partial_x + s\mu \tanh(\mu x)$, $L_{\mathrm{m}} = D_s^* D_s + A_{\mathrm{m}}$ and $D_s b = 0$. Its partner is

$$D_s D_s^* = -\partial_x^2 + s^2\mu^2 + s(1-s)\mu^2 \operatorname{sech}^2(\mu x) \ge s^2\mu^2.$$

The nonzero spectra of the two products agree; thus b is the only spatial bound profile, and its orthogonal space has the stated gap. The two retained frequencies need not both lie

below that gap. Normalization gives $N^2 = \mu\Gamma(s+1/2)/(\sqrt{\pi}\Gamma(s))$ and $\|b'\|^2 = s^2\mu^2/(2s+1)$, so its gradient cost is finite and retained in $A_{\rm m}$. Projection gives q_P exactly. The omitted block is $K_{QQ} = Q[L_{\rm m} + \mathrm{diag}(a^2, 0)]Q \geq m_{\rm gap}^2 Q$, and $K_{QP}\chi = Q(a\phi' + a^2 b q_\Theta, 0)$. Solving its equation gives $\Sigma(z) = K_{PQ}(K_{QQ} - z)^{-1}K_{QP}$ and the bound. □

The projected contact has profile ab and its complete square. The original k, K, I, J are recovered, rather than fitted after the projection. The remaining material field has not been reset or removed. For a response bound put

$$d_{\rm eff} = \|ab\|^2, \qquad \beta_a = 4\|a\|_\infty^2(1 + d_{\rm eff}/\alpha) + 2\|a^2 b\|^2/\alpha.$$

The inequality $\|\phi'\|^2 \leq 2(1 + d_{\rm eff}/\alpha)q_P$ yields $\Sigma(z) \leq \beta_a K_{PP}/(m_{\rm gap}^2 - z)$ as forms. For $\sigma > 0$, let $\epsilon = \beta_a/(m_{\rm gap}^2 + \sigma^2) < 1$, $R_{\rm ex} = P(K + \sigma^2)^{-1}P$ on the retained space and $R_P = (K_{PP} + \sigma^2)^{-1}$. Normalizing the Schur complement by $K_{PP} + \sigma^2$ gives $I - D$, $0 \leq D \leq \epsilon I$, hence

$$\|R_{\rm ex} - R_P\| \leq \frac{\epsilon}{(1-\epsilon)\sigma^2}.$$

This certifies the exponentially weighted causal response: $\int_0^\infty e^{-\sigma t}\sin(t\sqrt{K})/\sqrt{K}\, dt = (K + \sigma^2)^{-1}$.

For finite-time linear-response forcing $F(t) = K_{QP}\chi(t)$ and zero omitted response displacement and velocity, its exact reaction is

$$u_Q(t) = -\int_0^t K_{QQ}^{-1/2}\sin((t-r)\sqrt{K_{QQ}})F(r)\, dr, \qquad \sqrt{2E_Q(t)} \leq \int_0^t \|F(r)\| dr.$$

Here E_Q is the energy of that response, not its vacuum fluctuations. The force uses the actual retained history, including backreaction. An initial omitted excitation adds its homogeneous evolution. Near the material threshold, or near a retained resonance, a small projection alone does not bound unfiltered scattering or rapid gates. The full local Gaussian propagator retains its light cone; a single collective coordinate generally does not preserve that sharp front. The estimates concern linear response and omitted energy, not a complete quantum-state or interacting thermal equivalence. They give a finite-body reduction with an explicit remainder while the physical clock, nonlinear recoil and thermal renewal are still to be matched.

A.2 A local record drive on the same quantum material

A finite record can drive the material without replacing its vacuum by that of the projected oscillator. Keep the complete Gaussian operator K of Section A.1, let $\omega = \sqrt{K}$ and $H = d\Gamma(\omega)$, and use one common material vacuum calibration. For a real material smearing $h = (0, h_\Theta, h_Y)$ define

$$Q(h) = \frac{a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)}{\sqrt{2}}.$$

The bound $q \geq \alpha\|u\|^2$ gives $\langle h, K^{-1}h\rangle \leq \|h\|^2/\alpha$ by the inverse-form variational formula. Spectral Cauchy–Schwarz then gives $\|K^{-1/4}h\|^2 \leq \|h\|^2/\sqrt{\alpha}$. Thus these material smearings

are defined even when the carrier has no mass gap. No static infrared dressing unitary is assumed.

Let $f \in C_c^\infty(0,T)$ be real and $F_1 = \int |f|$. An acquired classical bit $r \in \{0,1\}$ selects the drive $-rf(t)Q(h)$. In the interaction picture its exact unitary is

$$U_{I,r} = e^{ir^2\Phi_h} W(rz_h), \qquad z_h = \frac{i}{\sqrt{2}}\omega^{-1/2}\int_0^T f(t)e^{it\omega}h\,dt,$$
$$\Phi_h = \frac{1}{2}\int_{s>t} f(s)f(t)\langle h, \omega^{-1}\sin((s-t)\omega)h\rangle\,dt\,ds. \tag{35}$$

The commutator is a scalar, so the Magnus expansion stops here. The Weyl convention and coherent-state overlap are standard [6]. In the Schrödinger picture the mean displacement is the full retarded response

$$\xi_h(t) = r\int_0^t \frac{\sin((t-s)\omega)}{\omega} h f(s)\,ds.$$

For centred Gaussian input, its added energy is

$$E_h = \|\omega^{1/2} r z_h\|^2 = \frac{r^2}{2}\left\|\int f(t)e^{it\omega}h\,dt\right\|^2 \le \frac{r^2F_1^2}{2}\|h\|^2.$$

It equals the actual force work $\int rf\langle h, \dot\xi_h\rangle dt$. The instantaneous forced form is bounded below by $-r^2 f(t)^2\langle h, K^{-1}h\rangle/2$. Adding the opposite scalar makes it nonnegative; that offset is not net work, heat or a construction of the driver.

To address the angle mode, choose smooth $0 \le \chi_R \le 1$, equal to one on $|x| \le R$ and zero outside a finite larger interval. Set

$$A_R = \int b^2\chi_R, \quad h_R = (0, b\chi_R/A_R, 0), \quad h_b = (0,b,0), \quad d_R = \|h_R - h_b\|.$$

Both have retained force $\langle b, h_\Theta\rangle = 1$. Only h_R is the permitted compact drive; h_b is a comparison smearing in the same full model. If $t_R = \int_{|x|>R} b^2$, then $d_R^2 \le t_R/(1-t_R)$ and $\|h_R\|^2 \le (1-t_R)^{-1}$. For $s = 1/2$, $t_R = 1 - (2/\pi)\arctan(\sinh(\mu R))$. Reducing this tail grows the actual drive support.

Proposition 24 (Local actuation with the initial state retained)**.** *Let $\Delta z = z_{h_R} - z_{h_b}$, with both drives evolved under full K. Then*

$$\|\Delta z\|^2 \le \frac{F_1^2 d_R^2}{2\sqrt{\alpha}}, \qquad \Delta E := \|\omega^{1/2}\Delta z\|^2 \le \frac{F_1^2 d_R^2}{2}.$$

For an actual initial field marginal σ_r with its reference retained, the half trace distance of the two complete outputs obeys

$$D_r \le \sqrt{1 - |\operatorname{Tr}\sigma_r W(r\Delta z)|^2}.$$

A classical distribution of records gives joint distance at most $\sum_r p_r D_r$. On the same Gaussian ground state this becomes $\sqrt{1-e^{-r^2\|\Delta z\|^2}}$. On a finite Dirichlet interval, for the actual full-K Gibbs state $\sigma_\beta = e^{-\beta H}/Z$, the two displaced field states satisfy

$$S(\sigma_{h_R}\|\sigma_{h_b}) = \beta r^2\Delta E, \qquad D_{\mathrm{field},r} \le \sqrt{\beta r^2 \Delta E/2}.$$

Proof. The bound on $K^{-1/4}$ and the spectral integral in Eq. (35) prove the first two inequalities. Writing $B_R = \int b^2 \chi_R^2 \le A_R$ gives $d_R^2 = B_R/A_R^2 - 1 \le A_R^{-1} - 1 \le t_R/(1-t_R)$. Purify the actual field and reference state. The overlap of the two outputs is its Weyl characteristic function, up to a scalar phase. The pure-state trace-distance formula and partial-trace contraction give the retained-reference bound. For the Gibbs identity, conjugate both outputs by $W(-rz_{h_b})$. Displacement preserves entropy, $\log \sigma_\beta = -\beta H - \log Z$, and the initial mean is zero. Its energy increment is $r^2 \Delta E$; Pinsker gives the field-only bound. □

The Gibbs energy bound concerns the reduced field states. A retained thermal purification still uses the characteristic-function bound; its covariance can make that distance larger. Here ΔE is the energy of the difference response, not $E_{h_R} - E_{h_b}$, which has a cross term. No independent ground state of K_{PP} is substituted. Its dynamic reduction still has the reaction error of Proposition 23. An actual old field state must use its own characteristic function; initial correlation is not replaced by a fresh thermal packet. Coherent record controls also retain the Magnus and Weyl phases and are not covered by the classical-record distance bound.

For a centred Gaussian input the covariance is unchanged. A quadratic source, renormalized with the same fixed vacuum subtraction in both experiments, therefore changes by the classical value $T[\xi_h]$. Nonzero initial mean adds its bilinear cross term. The local energy balance has force input $rf(t)h \cdot \dot{\xi}_u$; the driver must supply that work. It uses the complete response, including omitted material modes, not merely the isolated mode's energy. This is the fixed-texture Gaussian source, not the full moving quantum apparatus's stress. The emitted carrier is determined by that material motion. Its equation is $(\partial_t^2 - \partial_x^2)\xi_\phi = \partial_x(a\xi_\Theta)$. For zero past displacement, outside the compact contact the right and left outgoing profiles are

$$\xi_R(u) = -\tfrac{1}{2}\int a(y)\xi_\Theta(y, u+y)dy, \qquad \xi_L(v) = \tfrac{1}{2}\int a(y)\xi_\Theta(y, v-y)dy, \quad u = t-x,\ v = t+x.$$

Integrating the one-dimensional retarded Green function against $\partial_y(a\xi_\Theta)$ gives these endpoint integrals directly. The extra stresses of centred Gaussian input are $(\xi_R')^2$ and $(\xi_L')^2$. They use the complete material response, not an independent emitter. With no incoming drive-induced radiation, the total emitted carrier energy is at most the force work E_h; the remaining material energy is retained. A zero contact $a = 0$ emits no carrier source despite possible stored material energy. Opposite drives also have equal quadratic stress on centred input, so a command label alone does not identify the emitted source.

A finite menu of these record-selected unitaries admits the bounded prepared-bit dilation of Appendix I.1. For compact smooth h_R on the Gaussian ground state, its Weyl amplitudes have all positive frequency moments, so the required twice differentiable H graph orbits exist. Smoothing and autonomous return then use this same H. Other initial states need their own graph-moment certificate. The extra bit, clock, battery, program and retained outputs remain resources. The record must reach the entire drive support by physical transport before the operation; the dilation does not prove that transport or the clock's covariant stress. Approximating a smooth local source during autonomous execution also needs its relative source bound, not trace distance alone. The following estimate provides it for finite-interval output readings of this material.

A.3 A source certificate for the quantum material

Fix a Dirichlet interval of length L and keep all modes of the complete Gaussian K. Write its pointwise mass matrix as $mI \le M(x) \le M_{\max} I$, $m > 0$. For $X = (\phi, \Theta, Y)$ its spatial energy form is

$$q[X] = \|\phi' + a\Theta\|^2 + \|u'\|^2 + \langle u, Mu\rangle, \qquad u = (\Theta, Y).$$

With $C_g = 2 + 2\|a\|_\infty^2/m$ and $V = 2\|a\|_\infty^2 + M_{\max}$, Cauchy–Schwarz gives $\|X'\|^2 \le C_g q[X]$ and $q[X] \le 2\|X'\|^2 + V\|X\|^2$. Min–max on three Dirichlet components bounds the ordered frequencies:

$$c_- j \le \omega_j \le c_+ j, \qquad c_- = \frac{\pi}{3L\sqrt{C_g}}, \quad c_+ = \sqrt{2\pi^2/L^2 + V}.$$

The full tower, not a finite Fock cutoff, enters the next statement.

Let $\mathcal{Y} = (\Pi, \phi' + a\Theta, u', \sqrt{M}u)$ be the energy-factor fields, so their squared norm divided by two is the energy density. For a real symmetric bounded spatial matrix $B(x)$, $\|B(x)\| \le G$, and real $h \in C_c^\infty$, set

$$O = \frac{1}{2}\int h(t) \int :\mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x): \, dx\, dt.$$

Normal ordering uses the same K vacuum in both compared states. $B = gI$ tests the local energy. In a free exterior carrier region a rank-one choice tests $g(\Pi_\phi \mp \phi')^2/4$, the corresponding null stress. These tests may have negative quantum expectations.

Proposition 25 (A same-Hamiltonian quadratic-source bound)**.** *Put $H = d\Gamma(\omega)$, $H_N = \|h^{(N)}\|_1$, $N \ge 2$. The source is a Hermitian form on* $\operatorname{Dom} H^{1/2}$*,*

$$O = d\Gamma(A) + P + P^*, \qquad P = \sum_{ij} b_{ij} a_i a_j.$$

It obeys $|d\Gamma(A)| \le a_0 H$, $a_0 = G\|h\|_1$. Define

$$\kappa^2 = \sum_{ij} \frac{|b_{ij}|^2}{\omega_i \omega_j}, \qquad \nu^2 = \sum_{ij} \frac{|b_{ij}|^2}{\omega_i}.$$

Both converge, with the explicit sufficient bounds

$$\kappa \le \frac{GH_N}{2c_-^N}\sqrt{\zeta(2N-1) - \zeta(2N)},$$
$$\nu^2 \le \frac{G^2 H_N^2 c_+}{8c_-^{2N}}[\zeta(2N-2) - \zeta(2N-1)]. \tag{36}$$

For any reference energy $\Lambda > 0$, let $C_O = a_0 + 2\kappa + \nu/\sqrt{\Lambda}$. Then $(H+\Lambda)^{-1/2} O (H+\Lambda)^{-1/2}$ is a bounded Hermitian form of norm at most C_O.

Proof. Choose real eigenmodes e_j of K. If $L_s^* L_s = K$ is its spatial energy factor, the annihilation coefficients of $\mathcal{Y}$ are

$$v_j = (-i\sqrt{\omega_j/2}\, e_j, L_s e_j/\sqrt{2\omega_j}), \qquad \langle v_i, v_j\rangle = \omega_j \delta_{ij}.$$

Spatial multiplication by B is bounded by G on this factor space. Time evolution preserves the energy norm; integration against h proves the number-form estimate. The pair coefficient is $b_{ij} = \frac{1}{2}\int v_i^T B v_j dx\, \widehat{h}(-\omega_i - \omega_j)$, where $\widehat{h}(\xi) = \int h(t) e^{i\xi t} dt$. Integrating N times by parts gives

$$|b_{ij}| \le \frac{G\sqrt{\omega_i \omega_j} H_N}{2(\omega_i + \omega_j)^N}.$$

Summing over $i + j = k$ uses $k - 1$ pairs and $\sum_{j=1}^{k-1} j = k(k-1)/2$; the frequency bounds give Eq. (36).

For a unit finite-particle spectral vector of energy E, Cauchy–Schwarz applied to $\sqrt{\omega_i} a_i \psi$ and $\sum_j \overline{b_{ij}} a_j^* \psi/\sqrt{\omega_i}$ gives

$$|\langle P\rangle| \le \sqrt{E(\kappa^2 E + \nu^2)} \le \kappa E + \nu\sqrt{E}.$$

The creation-vector square is a vacuum term ν^2 plus a number form bounded by $\kappa^2 H$: the matrix $\omega^{-1/2} b \omega^{-1/2}$ has Hilbert–Schmidt norm κ. Thus $|\langle O\rangle| \le (a_0 + 2\kappa)E + 2\nu\sqrt{E} \le C_O(E + \Lambda)$. Pair tails converge in these two sums; the uniformly bounded number forms converge on the finite-energy core. Their normalized forms therefore extend to the stated bounded Hermitian form. □

This estimate concerns a spacetime-smeared source form, not an unsmeared density. Its constants depend on L; they do not prove a uniform real-line limit. Compact $W^{2,1}$ time tests also satisfy the $N = 2$ proof. No free chiral energy inequality has been assigned to the coupled material.

Corollary 26 (Output source error with references retained)**.** *For normalized actual and target vectors $\psi, \varphi \in \operatorname{Dom} H$, including their actual references, let $e = \|\psi - \varphi\|$ and $g_H = \|H(\psi - \varphi)\|$. Then*

$$|\langle O\rangle_\psi - \langle O\rangle_\varphi| \le C_O\sqrt{\Lambda e^2 + e g_H} \times \left[\sqrt{\Lambda + \|H\psi\|} + \sqrt{\Lambda + \|H\varphi\|}\right]. \tag{37}$$

Proof. Tensor the bounded normalized form with the reference identity and polarize. Its difference is bounded by C_O times $\|(H + \Lambda)^{1/2}(\psi - \varphi)\|$ times the sum of the two weighted norms. The first squared norm is at most $\Lambda e^2 + e g_H$; each remaining squared norm is at most $\Lambda + \|H\psi\|$ or $\Lambda + \|H\varphi\|$. □

The energy-regular return theorem supplies e, g_H for a specified same-H graph-domain operation. It consequently certifies these output source readings under common subsequent H evolution, not the missing driver or clock stress during its interaction. In fact its uniform physical rms bounds M_a, M_i suffice: $g_H \le M_a + M_i$ in Eq. (37). The resulting source error

tends to zero with the autonomous state error even as the history clock grows; its physical rms bounds do not grow with that clock. This does not add a permanent halt or decouple an always-running controller for free. A mixed input uses its retained purification. On this finite interval a full-K Gibbs purification has finite energy moments: with $q_\beta = e^{-\beta c_-}$,

$$\langle H\rangle \le \frac{c_+ q_\beta}{(1-q_\beta)^3}, \qquad \operatorname{Var} H \le \frac{c_+^2 q_\beta(1+q_\beta)}{(1-q_\beta)^5}.$$

The bosonic product partition function converges. Bounding occupations by $q_\beta^j/(1-q_\beta)$ and summing jq_β^j and $j^2 q_\beta^j$ proves the two estimates. All higher polynomial moments are finite as well. This does not prepare the Gibbs state, prove a particular gate's graph orbit, or renew a correlated reservoir.

For the vacuum-driven outputs $|z\rangle, |w\rangle$ of two actual forces on this same material, put $c = e^{-\|z-w\|^2/2}$, $E_z = \|\omega^{1/2} z\|^2$ and $\Delta E = \|\omega^{1/2}(z-w)\|^2$. Choosing only a comparison phase to make their overlap c gives the exact identity

$$\|(H+\Lambda)^{1/2}(|z\rangle - |w\rangle)\|^2 = c\Delta E + (1-c)(E_z + E_w + 2\Lambda).$$

Indeed $\langle z|H|w\rangle = \langle z|w\rangle\langle z, \omega w\rangle$. Their source difference is at most C_O times the square root of this expression times $\sqrt{E_z+\Lambda}+\sqrt{E_w+\Lambda}$. Proposition 24 bounds its two difference terms for compact forces. This comparison keeps full K; changing the box or using the projected oscillator changes the problem. For coherently controlled records the actual relative phases remain, rather than being physically aligned branch by branch.

Trace distance alone cannot replace these estimates: a trace-small admixture can have arbitrarily large energy. Nor does matching mean energy suffice. A small vacuum–two-particle amplitude changes a pair-source reading to first order while its energy changes to second order. The source certificate retains both the state and its energy norm. Matching this finite-interval Gaussian readout to a charged nonlinear continuum and the complete moving apparatus remains a separate physical construction.

B Finite return through one thermal field

The oscillator loop closes at one frequency. A propagating field has an interval of frequencies, and a sequence timed for its carrier does not close at every detuning. We now compute what remains, and how a single thermal line can be used over a finite adaptive protocol without resetting it between operations. The field theory, phase controls and reading instruments are still inputs.

B.1 Finite-width forces and the outgoing record

Let $\omega = \Omega + \delta$, $|\delta| \le B < \Omega$, and let g be real, even, square-integrable, and nonzero on an interval. In units $\hbar = 1$, the field Hamiltonian is $H_F = \int (\Omega+\delta) a_\delta^\dagger a_\delta \, d\delta$. For a system projector P with both eigenvalues $0, 1$, apply

$$H_s^I(t) = iP \int g(\delta) \left[u(t) e^{i\delta t} a_\delta^\dagger - \overline{u(t)} e^{-i\delta t} a_\delta \right] d\delta. \tag{38}$$

This is the interaction picture of a prescribed carrier-modulated force. It supplies two field quadratures and a clock; it is band-smeared, not compactly localized in space. At finite $u(t)$ the laboratory Hamiltonian is bounded below by $-|u(t)|^2 \int g^2/(\Omega+\delta)\, d\delta$. A scalar shift makes it nonnegative without altering the gate.

Use four disjoint rectangular pulses of width $\sigma < \tau$, centred at $j\tau$, with complex labels $z_j = (1, i, -1, -i)$ and heights $rz_j/(\sigma\sqrt{Z})$, where

$$Z = \int g(\delta)^2 \operatorname{sinc}^2(\delta\sigma/2) d\delta, \qquad f(\delta) = g(\delta)\operatorname{sinc}(\delta\sigma/2)/\sqrt{Z}, \quad \operatorname{sinc} x = \sin x/x.$$

Thus $\|f\|_2 = 1$. The individual field displacements are $\alpha_j = rz_j f e^{ij\delta\tau}$. Put

$$M(t) = \int |f|^2 \cos(\delta t) d\delta, \qquad C = 3M(\tau) - M(3\tau), \qquad F = rf(1 - e^{2i\delta\tau})(1 + ie^{i\delta\tau}).$$

Proposition 27 (Continuum loop and its thermal defect)**.** *The complete four-pulse unitary is*

$$U_{\text{loop}} = e^{ir^2 CP} D(PF), \qquad D(PF) = \exp \int P(Fa^\dagger - \overline{F}a) d\delta. \tag{39}$$

If $C > 0$, choose $r^2 = \theta/C$ for a target phase $e^{i\theta P}$, $\theta > 0$. For an initially independent thermal field with inverse temperature β, define

$$D_\beta = \int |F(\delta)|^2 \coth[\beta(\Omega + \delta)/2]\, d\delta. \tag{40}$$

The thermal state is specified by its Weyl characteristic; a continuum KMS state need not be a trace-class density matrix on vacuum Fock space. After removal of the target phase the system channel is dephasing between the two P blocks, with coherence factor $e^{-D_\beta/2}$. Its half diamond distance from the target unitary is $(1 - e^{-D_\beta/2})/2$, and its representation defect norm on a unit-norm off-block Pauli control is $1 - e^{-D_\beta}$.

No nonzero compact-time force through this single profile can close the continuum exactly. This last statement concerns the source (38)*, not all multimode control architectures.*

Proof. The source commutator is a scalar field commutator times $P^2 = P$, so the time-ordered exponential is a conditional displacement and a commuting P phase. Within each constant-phase pulse the self-phase vanishes after integration over the even band: its sine kernel is odd in δ. The six inter-pulse Weyl phases add to $r^2[3M(\tau) - M(3\tau)]$. Summing the four displacement vectors gives F. This derivation includes finite pulse width; the pulses are not delta kicks.

The thermal Weyl characteristic is $\langle D(F)\rangle_\beta = e^{-D_\beta/2}$. Tracing the field gives the stated channel, a mixture of identity and conjugation by $2P - I$ after the target phase. Orthogonal off-block superpositions attain the half diamond distance; multiplying its Heisenberg off-block factor gives the defect.

For a general integrable compact-time u the displacement is $g(\delta) \int u(t) e^{i\delta t} dt$. Exact unitary control after discarding the thermal field requires this displacement to vanish. The Fourier transform of compact-time u is entire. Vanishing on an interval where $g \neq 0$ forces it to vanish identically, and Fourier uniqueness gives $u = 0$. At one oscillator frequency only a single zero is required, which is why Proposition 12 has no such obstruction. □

For $\theta = \pi$ and $P = (I + X)/2$, the target is a corrective flip up to phase. With $\mu_2 = \int \delta^2 |f|^2 d\delta$,

$$C \geq 2 - \tfrac{3}{2}\tau^2 \mu_2, \tag{41}$$

$$\|F\|_2^2 \leq 8r^2\tau^2(1 + B\tau)\mu_2, \tag{42}$$

$$D_\beta \leq \coth[\beta(\Omega - B)/2]\|F\|_2^2. \tag{43}$$

The first uses $M(\tau) \geq 1 - \tau^2\mu_2/2$ and $M(3\tau) \leq 1$. For the second, $|1 - e^{2i\delta\tau}| \leq 2|\delta|\tau$ and $|1 + ie^{i\delta\tau}|^2 = 2 - 2\sin(\delta\tau) \leq 2(1 + B\tau)$. Temperature multiplies the residual record; it does not remove it.

B.2 One line throughout an adaptive sequence

Multiplying the reduced one-loop channels would assume a fresh field. That is not the evolution we consider. The same field and its correlations remain present, and subsequent projectors need not commute.

Proposition 28 (Finite reuse without independence)**.** *Start with one thermal line independent of system, controller and retained reference. A protocol uses at most L loops, interspersed with arbitrary system–controller instruments whose dilations do not act on the line. Projectors may be selected from acquired records and may differ between loops. Suppose every permitted branch has $D_\beta \leq D_{\max}$ and its target phase is implemented as in* (39)*. The complete actual process, including final records, system and field, is at operational state distance at most*

$$\min\{1, L\sqrt{D_{\max}}\}$$

from the process with ideal target gates and the same initial line. Every terminal loss in $[0, 1]$ has an expectation difference within this bound, as does the total variation of the complete record laws. No reset or independent-error hypothesis is used. Here operational distance is half the norm distance of the states on their joint observable algebra; it is half trace distance whenever a density-matrix description applies.

Proof. Purify the initial thermal state, or use its thermal GNS cyclic vector. For a loop acting on the ideal-prefix state,

$$\|(D(PF) - I)|\Psi\rangle\|^2 = 2\langle P\rangle(1 - e^{-D_\beta/2}) \leq D_\beta.$$

This is uniform over system inputs, retained entanglement and the record-controlled choice of projector. In the ideal prefix the line remains thermal and independent; the actual prefix need not. Telescope the actual unitary dilations against the ideal dilations, placing the ideal prefix on each difference and leaving the actual suffix as a norm-preserving factor. The vector difference is at most $L\sqrt{D_{\max}}$. The norm distance of vector states, followed by restriction to physical observables or processing records, gives the stated operational bound. The proof does not replace any actual correlated prefix by a reset state. □

For constant per-pulse relative amplitude error at most e_a, centre error d_t and carrier-phase error e_φ, with unchanged pulse widths and still disjoint pulses, each displacement changes in vacuum norm by at most

$$e = r[e_a + (1 + e_a)(Bd_t + e_\varphi)].$$

The six Weyl phases change by at most $6(2re + e^2)$. The same proof therefore gives the finite-setting bound

$$L\left[\sqrt{D_{\max}} + 4\sqrt{\coth[\beta(\Omega - B)/2]}\, e + 6(2re + e^2)\right]. \tag{44}$$

It concerns these specified setting errors, not arbitrary time-dependent source noise or heating added during the force.

As a concrete family take $g^2 = 15[1 - (\delta/B)^2]^2/(16B)$, $\Omega = 10$, $B = 1$, $\beta = .2$ and $\sigma = \tau/2$. The unfiltered second moment is $B^2/7$; since $\operatorname{sinc}^2(\delta\sigma/2) \le 1$, $\mu_2 \le B^2/(7Z)$ and $Z \ge \operatorname{sinc}^2(B\sigma/2)$ in the stated small-width range. At $\tau = 5 \times 10^{-5}$, $e_a = e_\varphi = 10^{-8}$ and $d_t = 10^{-9}$, conservative rational bounds in (44) put an entire 64-loop adaptive protocol below .00512 operational distance. Tight numerical integration gives one-loop error 1.48×10^{-9}. The sequence statement comes from the joint bound, not from treating those tiny errors as independent. The clock's frequency units are supplied by B; these numbers are not unlabelled laboratory seconds.

B.3 Source energy and control resources

For one loop and an independent zero-mean thermal input, the excess free-field energy is

$$\Delta E_F = \hbar\langle P\rangle \int (\Omega + \delta)|F(\delta)|^2 d\delta. \tag{45}$$

For the chiral current normalization of Paper B, its corresponding free outgoing excess stress is $4\langle P\rangle[\operatorname{Re}\psi_F(u)]^2$, where $\psi_F(u) = \int -i\sqrt{\omega/(4\pi)}F(\omega)e^{-i\omega u}d\omega$. The chosen g vanishes at band edges and gives finite first two time moments. This is the free field source, not the stress of its support, phase reference or switching apparatus.

A reused line has extra cross terms. Let $E_1 = \int \omega|F|^2 d\omega$ bound the unit-projector displacement energy of each loop, and let $n_{\max} = [e^{\beta(\Omega-B)} - 1]^{-1}$. For loops sharing the stated profile, with arbitrary projector choices,

$$|\Delta E_F^{(L)}| \le \hbar\left[2L\sqrt{(\Omega + B)n_{\max}E_1} + L^2E_1\right]. \tag{46}$$

Indeed the final annihilation field is its initial value plus $\sum_j F_j P_j^{\mathrm{past}}$, with $\|P_j^{\mathrm{past}}\| \le 1$. The quadratic term is at most L^2E_1 by the weighted L^2 triangle inequality. Each initial thermal cross term is at most $\sqrt{(\Omega + B)n_{\max}E_1}$ by Cauchy–Schwarz. Thus the single-loop energy cannot simply be multiplied by L. This bound prices an excess over the supplied thermal background, not that background itself.

Small final radiation does not mean small driving demands. With $\|g\|_2 = 1$, the four rectangular sources have

$$\int |u(t)|^2 dt = \frac{4r^2}{\sigma Z}, \qquad |H_{\min}| \le \frac{\hbar r^2}{\sigma^2 Z(\Omega - B)}$$

during a pulse. The first is a declared quadratic envelope charge, not heat; the second is the magnitude of a lower energy bound, not consumed work. Narrowing the loop reduces its

escaped record but increases these source and clock requirements. At the 64-loop settings the single-loop envelope charge is about 2.51×10^{5}, although its unit-projector radiated energy is about $4.49 \times 10^{-8}\hbar$. Finite pulse heights are allowed; ideal switching of the external apparatus is still a supplied control.

For fixed band, temperature and target phase, $D_\beta = O(\tau^2)$ as $\tau \to 0$. Taking $\sigma = \tau/2$ and choosing τ proportional to ε/L therefore suffices for joint error ε, with setting errors also reduced to order ε/L. Within this family the total quadratic envelope charge scales as L^2/ε. This is a constructive resource scaling, not a lower bound for every field-control architecture.

Finite reuse removes the independent-port premise; the initially thermal line is never reset. Stored records, controller renewal, exhaust and total source/geometric response still require a compatible apparatus, not merely a small reduced-channel error.

C Repeated renewal of a complete ring body

Use a metric-graph carrier with four complex chiral body fields on rings of length w, each coupled to its own thermal line over an arc $0 < \ell < w$. All spatial modes remain. Their charges are $(+1, -1, +1, -1)$; the first pair is the receiver, the second its current purifier. Two colocated, copropagating rails on each arc obey

$$i(\partial_t + \partial_x)\Psi = h(x)W(t-x)\Psi, \quad h \in C_c^\infty(0,\ell), \quad \int h = 1. \tag{47}$$

The exact ray transfer is $e^{-iW(u)}$, $u = t - x$. Take $W(u) = \theta(u)\sigma_y$ and include a physical ring connection with round-trip phase $e^{-i\Delta w}$. The overlap transit is included in w; the voltages and their switching work are supplied controls.

Choose $\tau = Kw$, $K \in \mathbb{N}$, a real $\lambda \in C_c^\infty(0,\tau)$, and $\theta = \sqrt{w}\lambda$, $|\theta| < \pi/4$. Iteration gives

$$A_w(u) = \prod_{j=0}^{K-1} \cos[\sqrt{w}\lambda(u + jw)], \qquad 0 \le u < w, \tag{48}$$

$$F_w(s) = \frac{\sin[\sqrt{w}\lambda(s)]}{\sqrt{w}} e^{i\Delta s} \prod_{j\ge 1:\, s+jw<\tau} \cos[\sqrt{w}\lambda(s + jw)]. \tag{49}$$

The old-ring multiplier is A_w; F_w is the captured zero-mode envelope. Put $\eta^2 = w^{-1}\int_0^w |A_w|^2$. Retaining the full complementary outgoing field gives $\|F_w\|_2^2 = 1 - \eta^2$ by CAR. Endpoint flatness makes the last product smooth. With $\Omega = 2\pi/w$, the captured nth mode has carrier $\Delta + n\Omega$.

The reference sea fills $n < 0$ and leaves $n \ge 0$ empty. Sum over all four species in

$$N_{\mathrm{ex}} = \sum_{n\ge 0} d_n^\dagger d_n + \sum_{n<0} d_n d_n^\dagger, \tag{50}$$

$$P = \Omega\left(\sum_{n\ge 0} n d_n^\dagger d_n + \sum_{n<0} |n| d_n d_n^\dagger\right), \qquad Q = N_{\mathrm{ex}} + P/\Omega. \tag{51}$$

Let $r(E)$ be the occupation of the normalized captured envelope with carrier E in the actual temperature-T KMS line. Its real envelope gives $r(-E) = 1 - r(E)$. For $A_w(w\xi) = \sum_k a_k e^{2\pi i k\xi}$ set

$$\alpha = \sum_k \sqrt{1+|k|}\,|a_k|, \tag{52}$$

$$V_A = \tfrac{1}{2}\sum_{k>0} k(k+1)|a_k|^2 + \tfrac{1}{2}\sum_{k<0} |k|(|k|+3)|a_k|^2, \tag{53}$$

$$S_Q = (1-\eta^2)\left[r(\Delta) + \sum_{n\geq 1}(1+n)\{r(\Delta+n\Omega) + r(n\Omega-\Delta)\}\right], \tag{54}$$

$$b = 2(\sqrt{V_A} + \sqrt{S_Q}). \tag{55}$$

Proposition 29 (Uniform renewal with retained history)**.** *Suppose $\langle Q\rangle < \infty$, $0 < \Delta < \Omega$, and the next incoming supports have their unconditional KMS marginals. Correlations with the body and all earlier records are allowed. The actual reset satisfies*

$$\sqrt{\langle Q\rangle_{\text{reset}}} \leq \alpha\sqrt{\langle Q\rangle_{\text{old}}} + b. \tag{56}$$

Between resets allow the translation-invariant pumps below and supplied logical noise on the four zero modes. For post-reset $x_j = \sqrt{\langle Q\rangle_j}$, if $\sqrt{2}\alpha < 1$, then

$$x_{j+1} \leq \sqrt{2}\alpha x_j + 2\alpha + b, \quad \sup_j x_j \leq \max\left\{x_0, \frac{2\alpha+b}{1-\sqrt{2}\alpha}\right\}. \tag{57}$$

Against the actual outside marginal, including all old references,

$$\tfrac{1}{2}\|\rho_{\text{body},R} - |\text{sea}\rangle\langle\text{sea}| \otimes \rho_R\|_1 \leq 2x_j. \tag{58}$$

At fixed cycle duration and T, one finite choice of pulse strength, gap and ring length makes this error arbitrarily small uniformly in the number of cycles, with four active outlet species.

Proof. Apply all weighted wrong-occupation annihilators to a purification of the actual joint state. Weighted convolution, using $1 + |n| \leq (1+|m|)(1+|n-m|)$, bounds their old excitation-annihilation part by $\alpha\sqrt{\langle Q\rangle}$. CAR bounds the creation covariance by the identity; its squared norm is at most V_A per species. The incoming vector has squared norm S_Q. Minkowski's inequality on the four-species direct sum proves (56), without factoring the old state from the line. Pumps and logical noise preserve P, and $N_{\text{ex}} \leq P/\Omega + 4$, so $Q_{\text{after gate}} \leq 2Q_{\text{before gate}} + 4$ in expectation. This gives (57). The nonsea projection is bounded by $N_{\text{ex}} \leq Q$. Projection of a joint purification onto the pure sea and trace-distance contraction give (58).

The full sums are finite before imposing any numerical cutoff. Set $\Gamma = \int \lambda^2$ and $g_w = \log\cos(\sqrt{w}\lambda)$, extended by zero. For each $m \geq 3$, $G_m = \sup_{w\leq w_0} w^{-1}\|g_w^{(m)}\|_1 < \infty$ if $\sqrt{w_0}\|\lambda\|_\infty < \pi/4$. Poisson summation bounds the nonconstant Fourier coefficients of $\log A_w$ by $G_m w^m/(2\pi|k|)^m$. Weighted Wiener exponentiation therefore gives

$$\alpha \leq e^{-\Gamma/2+b_m}, \quad V_A \leq 2e^{-\Gamma}(e^{b_m}-1)^2, \quad b_m = \frac{2G_m w^m}{(2\pi)^m}[\zeta(m) + \zeta(m-1)]. \tag{59}$$

Write $F_w = e^{i\Delta s} f_w$. Every derivative of its logarithmic product is a sum of $O(1/w)$ terms of size $O(w)$, with flat endpoints. Thus $D_m = \sup_{w \le w_0} \|f_w^{(m)}\|_2^2 < \infty$. Splitting the Fourier integral at $E/2$ bounds the exact Fermi occupation by

$$r(E) \le e^{-E/(2T)} + \frac{(2/E)^{2m} D_m}{1-\eta^2}, \qquad E > 0. \tag{60}$$

For $\Omega \ge 2\Delta$ and $z = e^{-\Omega/(4T)}$,

$$\begin{aligned} S_Q &\le (1-\eta^2) r(\Delta) + 2(1-\eta^2) \frac{z(2-z)}{(1-z)^2} \\ &\quad + 2D_m (4/\Omega)^{2m} [\zeta(2m) + \zeta(2m-1)]. \end{aligned} \tag{61}$$

Choose Γ large, then Δ large, then $w = \tau/K$ small. All errors become small simultaneously. A finite initial reset puts x_0 at the same tolerance. Earlier nonselective even operations on disjoint line characteristic supports leave the next unconditional KMS marginal unchanged; guards longer than the overlap transit provide that separation. □

The mixed target is restored by a local neutral quartet pump

$$V_4 = \int_0^w (D_1^\dagger D_2^\dagger D_3^\dagger D_4^\dagger + \text{h.c.})\, dx. \tag{62}$$

It commutes with P. On the sea, creation requires four nonnegative mode indices summing to zero: only the four zero modes occur. The reverse transition obeys the same argument. Their two-dimensional subspace is exactly invariant, with matrix element $1/w$.

For the full operator, bosonize the periodic fields with $q_i \in \mathbb{Z}+1/2$. The charge coordinate $\phi_c = \sum_i \phi_i/2$ gives the weight-two vertex $e^{2i\phi_c}$. At fixed transverse charges, $q_c = q_0 + 2\mathbb{Z}$ with $q_0 \in \frac{1}{2}\mathbb{Z}$, a fixed parity sector of a charge fermion; the other three bosons and the original Klein phases remain. The weight-two vertex splits this stress into two positive $c = 1/2$ Virasoro stresses with velocities $1 \pm \kappa/\pi$ [1]. The four Ramond ground weights sum to $1/2$, hence

$$\|V_4 \psi\| \le \pi^{-1} \|(P + \Omega/2)\psi\|. \tag{63}$$

Finite-dimensional P eigenspaces are invariant; their direct sum defines the self-adjoint pump. The pair pump $B = \int (D_1^\dagger D_2^\dagger + \text{h.c.}) dx$ obeys $|B| \le P/\Omega + 1$: every active nonzero pair costs at least Ω in P. For sea-normal-ordered number $N \ge -P/\Omega$,

$$P + \Delta N + \kappa V_4 + b_2 B \ge \left(1 - \frac{\Delta}{\Omega} - \frac{|\kappa|}{\pi} - \frac{|b_2|}{\Omega}\right) P - \frac{|\kappa|}{w} - |b_2|. \tag{64}$$

Small enough pump peaks give a common finite physical ground offset and positive Hamiltonian. This offset is part of the apparatus source.

A quartet pulse with phase $e^{-4i\Delta t}$ and area $w^{-1} \int \kappa(t) dt = \arcsin \sqrt{\epsilon}$ prepares

$$|\Phi_\epsilon\rangle = \sqrt{1-\epsilon}\, |00\rangle_A |00\rangle_R + \sqrt{\epsilon}\, |11\rangle_A |11\rangle_R. \tag{65}$$

The pair pump selects logical Z or X, retaining spectrum $(1-\epsilon, \epsilon)$ and the actual current purifier. Unitary contraction preserves (58); the next reset exports that purifier and its

correlations through the same lines. Fixed gate duration requires $\kappa_{\mathrm{peak}} = O(w)$, compatible with (64) and the ordered reset parameters.

The body has finite energy-constrained storage capacity rather than finite Hilbert dimension. Its positive-energy Gibbs partition function is finite for every inverse temperature $\beta > 0$, giving $S \le \beta\langle H\rangle + \log Z(\beta)$. Four complex outlets, any anomaly-cancelling partners, gap energy, drive work and stored outputs are charged. Fixed per-cycle error does not imply vanishing error for the whole growing history; that stronger task requires decreasing error per cycle and increasing resources. Autonomous covariant driving is treated below. Direct renewal exports the current purifier's two correlated occupation records into different species. Against a thermal reference that factorizes across species, their redundancy gives a positive complete-output relative-entropy rate whenever the reset and full-joint clock errors are small enough [23, Proposition "Output deficit and direct renewal"]. Improving this reset alone therefore does not approach the sharp heat-current limit. That task requires a physical output converter with its own work, memory and charge account.

Proposition 30 (Neutral block compression by physical gates)**.** *Buffer L original noisy A/R quartets in L equal-circumference slots of each physical charge species $(+,-,+,-)$. Permit local current phases, the pair and quartet pumps above, and same-charge quadratic mixers between connected slots. These prescribed gates can approximate a reversible code which confines their 4^L-dimensional support to m_L active zero modes and fixes the remaining modes, retaining arbitrary old references and intercycle correlations, where*

$$m_L = \min\left\{ r : \binom{r}{\lfloor r/2\rfloor} \ge 4^L, \quad 0 \le 2L - \lfloor r/2\rfloor \le 4L - r \right\} = 2L + \tfrac{1}{2}\log_2 L + O(1). \quad (66)$$

No additional pure working register or outgoing species is used. The buffer, initialization, control, offsets and eventual discharge are physical resources.

Proof. Relabel only the existing negative-charge zero modes by $f_i = d_i^\dagger$ and the positive-charge ones by $f_i = d_i$. Neutrality is $\sum f_i^\dagger f_i = 2L$ on $4L$ modes; this relabeling leaves the full sea and original charge representation unchanged. Pair pumps and same-charge mixers become one-body rotations. Connected rotations and currents generate $U(4L)$. The quartet becomes, with its calibrated Klein phase, $Q_{ab;cd} = f_a^\dagger f_b^\dagger f_c f_d + \mathrm{h.c.}$. Permutations move it to any four distinct labels.

Set $n_i = f_i^\dagger f_i$. Balanced rotations in $(a,c),(b,d)$, followed by averaging over their sixteen current sign phases, give $-\frac{1}{2}(n_a - n_c)(n_b - n_d)$. This average is implemented by a deterministic Trotter word, without a discarded random record. Quadrilateral differences span all two-body diagonal terms modulo currents on the fixed-number sector: their missing coefficient arrays are $\beta_{ij} = u_i + u_j$, for which $\sum_{i<j}\beta_{ij} n_i n_j = (2L-1)\sum_i u_i n_i$. Choose a diagonal $H_D = \sum \alpha_i n_i + \sum \beta_{ij} n_i n_j$ whose finitely many hopping transition frequencies are nonzero and distinct up to sign. Such coefficients avoid finitely many proper hyperplanes. Vandermonde interpolation in iterated commutators with H_D isolates the real and imaginary generators of each individual transition between occupied $2L$-subsets. These subsets form a connected Johnson graph. Its edge generators give the full special unitary group on the neutral sector; finite group-commutator and Trotter words approximate each required Givens rotation to the stated tolerance.

Enumerate the original 4^L pair-code basis states and inject them into the first 4^L occupation strings on m_L active modes with $\lfloor m_L/2 \rfloor$ particles. Fix the inactive pattern to $2L - \lfloor m_L/2 \rfloor$ particles and extend the injection to a neutral-sector unitary. Its action is $U_L \otimes I$ on every old reference. Stirling's formula gives (66). For the physical embedding, the translation-invariant equal-ring integrals commute with P. Its complete $P = 0$ sector is precisely the zero-mode Fock space tensored with the same spectator sea; the control word never projects out the full field tower. The previous relative-energy margins, with finite offsets and small primitive amplitudes, define the full-field gates. Unitary contraction carries a certified input-reset error through the word. □

For example, two quartets occupy eight zero modes but need only six active modes after this encoding. The fixed occupations result from reversible redistribution, not from a recurring purity supply. Two paid banks can hold old batches while new states arrive; both must subsequently be discharged and reused through the same four lines. The compiler may be very long and expensive. It proves finite prescribed controllability, not a sustainable autonomous clock or an efficient fixed-rate layout. It also preserves the data spectrum: concentrating it does not prepare thermal occupations of the outlet translation generator. The next two propositions supply finite physical loading, sorting and return; their clock, source and repeated-inlet accounts remain distinct.

Proposition 31 (A finite-width packet interface)**.** *Fix a physical charge species, m paid equal-ring memories and one of its original outlet lines. Let $v(t)$ be a smooth row of m overlapping envelopes, flat outside an interior guard of $(0, \tau)$, with $\int v^\dagger v \, dt = (1 - \eta^2) I_m$, $0 < \eta < 1$. A local Hermitian mixer on a finite common arc emits these memory columns with $O(w)$ packet error and terminal amplitude $\eta I_m + O(w)$ as the ring round-trip time w decreases. An analogous mixer captures any such incoming overlapping packet span. All original body modes and the complementary outgoing field remain in the canonical map.*

For a finite physical carrier of length V with no return during the window, let the joint input have physical excitation energy at most E, retaining arbitrary old references. With $\Omega = 2\pi/w$, $M_\Lambda = O(V\Lambda) + m$ low modes and any fixed sufficiently large smoothness order r, the actual and comparison full outputs have half trace distance at most

$$C\left(\sqrt{E/\Lambda} + \sqrt{E/\Omega}\right) + C_{m,V,\tau,\eta,r}\left[(M_\Lambda + 1) w (1+\Lambda)^2 + (1+\Lambda)^{1-r} + w^{r-1}\right]. \tag{67}$$

The constants include the supplied envelope norms. First choose Λ, then w. This is a finite-carrier, prescribed-control result; it is not a global trace theorem for an infinite KMS rail.

Proof. Set $K(t) = I - \int_0^t v^\dagger v \, ds$, $\Phi = K^{1/2}$ and

$$\lambda^\dagger = v\Phi^{-1}, \qquad A = \dot{\Phi}\Phi^{-1}, \qquad H = i\left(A + \tfrac{1}{2}\lambda\lambda^\dagger\right).$$

Since $K \geq \eta^2 I$ and $A + A^\dagger = \Phi^{-1}\dot{K}\Phi^{-1} = -\lambda\lambda^\dagger$, H is smooth and Hermitian. The comparison equations

$$\dot{d} = Ad - \lambda\psi_{\text{in}}, \qquad \psi_{\text{out}} = \psi_{\text{in}} + \lambda^\dagger d$$

have homogeneous propagator $\Phi(t)$ and emitted columns $\lambda^\dagger\Phi = v$. For capture, choose a column F with $\int FF^\dagger = (1-\eta^2)I$, and use

$$R(t) = \left[I - \int_t^\tau FF^\dagger\, ds\right]^{1/2}, \quad \lambda = R^{-1}F, \quad A = -R^{-1}\dot{R}.$$

The same identity gives Hermitian H; the future propagator is $G(\tau,t) = R(t)$, so the capture column is $-F(t)$. Its sign is an ordinary current phase, not reversed chiral transport.

In units of propagation speed one, choose a smooth arc profile $h_w(x) = w^{-1}h_0(x/w)$ of integral one and use the actual local potential

$$\mathcal{F}_w(t,x) = h_w(x)W_w(t-x), \qquad W_w = \begin{pmatrix} wH & -i\sqrt{w}\lambda \\ i\sqrt{w}\lambda^\dagger & 0 \end{pmatrix}. \tag{68}$$

The internal gap is lowered for this window, with its work charged. Along a characteristic the exact arc transfer is $S_w = e^{-iW_w} = \left(\begin{smallmatrix} B_w & C_w \\ D_w & E_w \end{smallmatrix}\right)$. Thus the full fields obey

$$b(t) = B_w(t-w)b(t-w) + C_w(t-w)\psi_{\mathrm{in}}(t-w), \qquad \psi_{\mathrm{out}}(t) = D_w(t)b(t) + E_w(t)\psi_{\mathrm{in}}(t).$$

In each fixed derivative norm, $B_w = I + wA + O(w^2)$, $C_w = -\sqrt{w}\lambda + O(w^{3/2})$ and $D_w = \sqrt{w}\lambda^\dagger + O(w^{3/2})$. Telescoping the noncommuting ordered products over τ/w returns gives the $O(w)$ comparison. The complete star is unitary, so its emitted Gram matrix plus its residual-body Gram matrix is exactly I_m.

The full tower also contributes. Each positive derivative of B_w is $O(w)$; differentiating an ordered product sums $O((\tau/w)^j)$ terms with j differentiated factors, each supplying $O(w^j)$. Its derivatives are therefore bounded uniformly in small w. Flat guards make the residual multiplier periodic on the ring. Its nonconstant Fourier coefficients satisfy $\|a_k\|_{\mathrm{HS}} \le C_r w^r/|k|^r$. The emitted and captured envelopes have the corresponding uniform smooth bounds; their nth body aliases retain the exact factors $e^{\pm in\Omega t}$.

The complementary line kernel is the direct term $E_w(t)\delta(t-s)$ plus its causal delayed returns

$$\sum_{j\ge1} D_w(t)B_w(t-w)\cdots B_w(t-(j-1)w)C_w(t-jw)\delta(s-t+jw).$$

Each coefficient and each fixed center derivative is $O(w)$, with $O(\tau/w)$ terms. The center derivative $\partial_t + \partial_s$ leaves both these delayed diagonals and the comparison Volterra factor $\Theta(t-s)$ unchanged. Fourier integration by parts therefore bounds its nonidentity kernel by $C_r(1+|\omega-\omega'|)^{-r}$, uniformly in w. Across opposite sea polarizations, $|\omega-\omega'| = |\omega| + |\omega'|$; the squared tail above Λ is $O((1+\Lambda)^{2-2r})$. On a fixed low-frequency subspace, the causal sum is a Riemann approximation with one-particle error $Cw(1+\Lambda)^2$. Together with the body alias bounds this controls the relative transformation's opposite-polarization Hilbert–Schmidt block.

For the state estimate, a proof-only energy cutoff excludes field excitations above Λ and nonzero body excitations with probability at most $E/\Lambda + E/\Omega$. Gentle projection works with the old reference retained. The remaining finite excitation space has M_Λ modes, so its Fock lift costs at most M_Λ times the one-particle error. The canonical pair-rotation decomposition bounds the vacuum-lift error by the opposite-polarization Hilbert–Schmidt

norm [48, Section 7.4, Theorem 31]. The actual normal-ordered lifts may differ by an overall phase, irrelevant for this prescribed channel. Combining the cutoff, low-band and full-tower estimates gives (67). □

Capture retains the actual covariance $C_{\mathrm{cap}} = \int F_w(t) C_{\mathrm{in}}(t,s) F_w(s)^\dagger \, dt\, ds$, including off-diagonal packet correlations and old body–line terms. Orthogonal wavefunctions need not have independent thermal occupations. A quantum program coherently selecting different loaders also requires their actual relative lift phases to be calibrated and charged. For the same finite graph, the source comes from its full covariance and local point splitting, and the work is $\int \langle \partial_t H_{\mathrm{graph}} \rangle dt$, including gap changes, contact energy and physical ground offsets. With $C_h = \int h_0^2$, the local quadratic scale satisfies

$$\int \operatorname{tr} \mathcal{F}_w^2 dx = C_h \left[w \operatorname{tr} H^2 + 2\|\lambda\|^2 \right].$$

This does not bound all higher spatial derivatives or clock resources uniformly. Nor does (67) alone control heat: an old spectator occupied with probability E/Ω can export energy E while disappearing in trace norm. Heat or source convergence requires propagated uniform energy integrability, such as a controlled higher energy moment or explicit weighted output tails. The packet map supplies the loading step for the following thermal converter. Its energy estimate uses the actual carrier rather than packet occupation numbers assigned independently of it.

Proposition 32 (Charge-sector thermal sorting on a retained carrier)**.** *Use the original four charge species and neutral gates, and retain four physical carrier loops of round-trip time $V = L/\nu$ in their complete finite-volume Gibbs state $\tau_{T,V}$, $T > 0$. Include its purification and all old references. Supply L independently realized noisy original quartets with marginal $\rho_S = \rho_0^{\otimes L}$, where*

$$\rho_0 = \begin{pmatrix} (1-q)(1-\epsilon) & 0 & 0 & (1-2q)\sqrt{\epsilon(1-\epsilon)} \\ 0 & q\epsilon & 0 & 0 \\ 0 & 0 & q(1-\epsilon) & 0 \\ (1-2q)\sqrt{\epsilon(1-\epsilon)} & 0 & 0 & (1-q)\epsilon \end{pmatrix}, \qquad h_\lambda = S(\rho_0).$$

The source and complete carrier initially have a product marginal; their external purifications are retained. There are prescribed finite words of the original smooth gates that return the source near its original blank and achieve, in natural units,

$$\lim_{\Lambda\to\infty} \limsup_{L\to\infty} \frac{D(\rho_{B,\mathrm{out}} \| \tau_{T_*,V})}{V} = 0, \qquad T_* = T + \frac{6\nu h_\lambda}{\pi c_{\mathrm{ex}}}, \quad c_{\mathrm{ex}} = 4. \tag{69}$$

Implementation error can be made arbitrarily small after choosing each finite block and cutoff Λ, with physical energy tails controlled. The entire output B, its uncaptured complement and every atypical label are present in this relative entropy. Finite paid internal pipeline stages permit the specified throughput without adding external outlets. This is a finite-carrier converter; an actual stationary inlet, repeated preparation and an autonomous reused controller require further physical constructions.

Proof. On each loop let $D = -i\partial_x$, P_Λ its Fourier subspace with $|\omega| \le \Lambda$, and χ a smooth cutoff flat zero in endpoint guards of length g. If $P_\Lambda \chi^2 P_\Lambda u_j = g_j u_j$, retain $g_j \ge 1 - e$ and set $f_j = \chi u_j / \sqrt{g_j}$. These are orthonormal smooth envelopes for Proposition 31. For their isometry F, the actual captured matrices are

$$C_T = F^*(1 + e^{D/T})^{-1} F, \qquad h_F = F^* D F.$$

They need not commute. A physical one-body rotation diagonalizes C_T for its spectral probabilities, whereas the final occupation states use the eigenbasis of h_F. Relabel existing negative-charge modes by $f = d^\dagger$ as in the neutral code, retaining their energy $-h_F^T$ and the physical sea constant.

Here is an exact finite count. In total charge sector Q, let $E_{\beta,Q}, S_{\beta,Q}, \sigma^2_{\beta,Q}$ be the energy, entropy and energy variance of the actual carrier-sector Gibbs matrix. For $b_Q > \sigma_{\beta,Q}$ there are at least

$$\left(1 - \frac{\sigma^2_{\beta,Q}}{b_Q^2}\right) e^{S_{\beta,Q} - \beta b_Q} \tag{70}$$

occupation states below $E_{\beta,Q} + b_Q$. Chebyshev gives this shell's mass; each state in it has probability at most $e^{-S_{\beta,Q} + \beta b_Q}$. Map the charge-resolved good input eigenlabels injectively into these low-energy states with the source blank, then complete the injection to a sector bijection. Every bad label occupies a remaining physical state. If their probability is p_{bad},

$$\Pr(S_{\mathrm{out}} \ne b) \le p_{\mathrm{bad}}, \qquad S(S_{\mathrm{out}}) \le h(p_{\mathrm{bad}}) + 4L p_{\mathrm{bad}} \log 2.$$

For the depolarizing source specified in the proposition, exact blank restoration would violate the full-rank finite input's rank count. No label is postselected away.

At fixed sufficiently large Λ, the source eigenvalue strings and the Gaussian carrier's actual spectral probabilities have information variance $O(L)$. An information margin $L^{9/10}$ and charge window $|Q| \le L^{5/8}$ lose $o(1)$ probability. Choose the finite-bank target entropy larger than the input entropy by $3L^{19/20}$ and put $b_Q = L^{17/20}$. For these charges exponential tilting of the bounded occupation Bernoullis gives $\log \Pr_\beta(Q) = -O(Q^2/L + \log L)$. Fourier inversion bounds the tilted probability at its mean below by $cL^{-1/2}$: its characteristic function is bounded by $e^{-cL\sin^2(\theta/2)}$, and its scaled central part converges to a nonzero Gaussian integral. Conditioning its $O(L)$ energy variance therefore gives $\sigma^2_{\beta,Q} = O(L^{3/2})$ and $|E_{\beta,Q} - E_\beta| = O(|Q| + L^{3/4})$. The count loss in (70) is smaller than the entropy margin. Hence $p_{\mathrm{bad}} \to 0$, source entropy is $o(L)$, and the carrier energy excess is $o(L)$.

The bijection uses the original gates. For a hop $a \to b$, a quadrilateral interaction $(n_a - n_c)(n_b - n_d)$ has difference $n_d - n_c$. Two distinct spectator subsets have the same differences for every c, d only when one is empty and the other full. Generic current and quadrilateral coefficients therefore separate all other oriented edges; finite Vandermonde filtering and the connected occupation graphs give independent special-unitary actions in sectors $2 \le k \le m - 2$. The linked $k = 1, m - 1$ sectors are unnecessary: the original source has $k_S = 2L$ on $4L$ modes, so every occupied source–carrier sector is interior. Spectral diagonalization and the chosen bijection are finite commutator/Givens words. Their phases are retained.

Capture and return act on the same complete carrier as $U^*_{\mathrm{load}} U_{\mathrm{sort}} U_{\mathrm{load}}$. For commensurate loops $U_0(V) = 1$ up to its declared sea phase. Thus the inverse of a free interval δ is the

positive wait $V-\delta$. Smooth signed current kicks invert the local mixing steps: evolution with $H_0+(a/\varepsilon)J(F)$ for time ε approaches $e^{-iaJ(F)}$ on the smooth energy core, with error at most $\varepsilon\int_0^1 \|H_0e^{-isaJ(F)}\psi\|ds$. Higher weighted versions follow from smooth multiplication and the summable opposite-sea tails. Reverse a finite Trotter word using these waits and kicks. Their work, peak ground shifts, relative phases and contact sources are paid. Its G whole-loop waits require G internal storage stages; narrowing a contact does not remove that latency. All stages have the original species and only the original four external outlets.

To pass from the bank to the complete physical carrier, let d be the number of discarded Fourier vectors and M the bank size. Then $d\le C(\Lambda g+1)/e$. The retained Fourier span need not be invariant under D; its discarded cross block contributes $C\Lambda d$, in addition to the cutoff error $CM(\Lambda\sqrt{e}+1/g)$. Taking $g=V^{1/2}$ and $e=V^{-1/4}$ makes these and the entropy compression error $o(V)$ at fixed Λ. The unselected thermal tails have energy and entropy per V at most $C(1+\Lambda)^2e^{-\Lambda/T_*}$. The ideal conjugation preserves the entropy of source plus full carrier, so Araki–Lieb and subadditivity give

$$|S(B_{\mathrm{out}})-S(\tau_{T,V})-Lh_\lambda|\le S(S_{\mathrm{out}}).$$

Together with the physical energy estimate and Gibbs identity, this bounds the deficit in (69) by $o_\Lambda(1)+C(1+\Lambda)^2e^{-\Lambda/T_*}$. Full finite-carrier Gibbs moments and the smooth gate graphs control implementation tails; Gibbs energy-constrained entropy continuity then gives the stipulated finite accuracy. Trace error alone would not give this energy or entropy conclusion. □

Remark 33 (Other supplied neutral source spectra). The proof of Proposition 32 also applies to a product source $\bigotimes_{j=1}^{L}\sigma_j$ from a fixed finite alphabet of known states in the same neutral quartet code, with the same joint source–carrier product marginal, paid gates and controlled physical energy tails. Joint operations include the receiver and its current, resettable purifier; external old references remain untouched. Put $s_L=\sum_j S(\sigma_j)$ and suppose $s_L/L\to s$ and the source information variance is $O(L)$. Replace h_λ by s in T_* and Lh_λ by s_L in the entropy identity. The finite-bank target uses the actual s_L plus the same subextensive entropy margin; its target temperature tends to T_*. No rate of convergence of s_L/L is needed. Eigenlabel concentration uses this variance and entropy. The source still has $k_S=2L$, so the neutral-sector compiler remains in its interior sectors. The final full-carrier entropy bound uses s_L; thermal counts, energy tails and paid loading/unloading are unchanged. HAFF-D's pointer-aligned dephasing supplies such rank-two sources at fixed noise parameters [23]. This is supplied-state renewal, not recovery of arbitrary unknown content against a forbidden purifier or a proof of the product marginal for an adaptive source.

The source debit is the complete quartet entropy h_λ. For $0<q,\epsilon<1/2$, its difference from the bare receiver change is the correlation lost in the logical depolarizing step:

$$h_\lambda-[h(\epsilon_q)-h(\epsilon)]=I(A:R)_{\mathrm{before}}-I(A:R)_{\mathrm{after}}\ge 4q\,h(\epsilon)>0.$$

The equality uses the pure corrected logical quartet and its noisy marginals; joint convexity for the depolarizing mixture gives the bound. With a conserved classical axis P, the source entropy conditional on P is Lh_λ. Its near-blank return removes $s^{PS}=Lh_\lambda-o(L)$ from

the complete source–program system. Entropy conservation with the independent thermal carrier gives

$$S(B_{\mathrm{out}}|P) - S(\tau_{T,V}) = s^{PS} + I(S_{\mathrm{out}} : B_{\mathrm{out}}|P).$$

No noise purifier is granted as usable side information. An autonomous controller's sensing and renewal must also enter its complete ledger. HAFF-D gives the corresponding conditional deficit account [23].

The fixed Z and X noisy quartets are isospectral. Their entropy-optimal outlet heat agrees to leading order, although their retained correlations and recovered work can differ. The direct source contrast below belongs to its specified unsorted reset. Its causal intervention remains valid, but does not certify persistent heat gain for the sharp converter.

Corollary 34 (The selected axis changes the maintenance source)**.** *Fix $0 < \epsilon < 1/2$ and $0 \le q < 1/2$. On the same complete post-reset state, compare the two physical axis preparations above, followed by the logical depolarizing channel with $\epsilon_q = \epsilon + q(1-2\epsilon)$ and the same reset. Do not undo the X rotation. There are fixed finite reset parameters for which, uniformly after the initial preparation cycles,*

$$\int \zeta(t)(\langle T_{\mathrm{out}}\rangle_X - \langle T_{\mathrm{out}}\rangle_Z)dt \ge \frac{\Delta(1-\eta^2)}{2}(1-2q)(1-2\epsilon) > 0, \tag{71}$$

where $\zeta \ge 0$ is smooth and equals one on the emitted pulse. This is a paired causal intervention on one pre-preparation state, not a comparison of histories selected by conditioning on different acquired outcomes.

Proof. The ideal total zero occupations are $2(\epsilon_q + \epsilon)$ and $1 + 2\epsilon$. Their difference is $D = (1-2q)(1-2\epsilon)$. Let v_w be the real emitted zero-mode envelope. The emitted nth mode is $v_n = e^{-i(\Delta+n\Omega)t}v_w$, and $J_k = \int \zeta v_w^2 e^{-ik\Omega t}$, $J_0 = 1-\eta^2$. The exact old-body matrix of the smeared stress is

$$B_{mn} = [\Delta + \Omega(m+n)/2]J_{n-m}.$$

Its diagonal is $J_0(P + \Delta N_s + \Delta N_0)$ after sea normal ordering. Both interventions preserve P; $|N_s| \le P/\Omega$ and $\|N_0\| = 4$. If d bounds the half-trace reset error and M_E bounds $\langle P/\Omega\rangle$, their diagonal contrast is at least $\Delta J_0(D - 16d - 2M_E)$.

These errors can be controlled independently of the zero occupation. Separating the old zero mode in the excitation-vector proof gives

$$\sqrt{\langle P/\Omega\rangle_{j+1}} \le \alpha\sqrt{\langle P/\Omega\rangle_j} + 2(\sqrt{Z_A} + \sqrt{V_P} + \sqrt{S_P}),$$

where $Z_A = \sum_{n\neq 0}|n||a_n|^2$, $V_P \le V_A$ is the nonzero-mode creation leakage, and S_P is S_Q with its zero term removed and weight n instead of $1+n$. Thus $M_E \to 0$ as $w \to 0$ at fixed Γ, Δ. Before reset take $M_Q = 4 + 2M_E$ as a bound on $\langle Q\rangle$.

Set $R_2 = \sum_{k\neq 0}(1+|k|)^2|J_k|$. Weighted Schur estimates for equal sea polarizations and a Hilbert–Schmidt estimate across the sea give a two-branch off-diagonal error at most $2(\Delta+\Omega)R_2(M_Q + 4\sqrt{M_Q})$. For $m \ge 5$, integration by parts gives

$$R_2 \le 2\|(\zeta v_w^2)^{(m)}\|_1\Omega^{-m}[\zeta_R(m) + 2\zeta_R(m-1) + \zeta_R(m-2)],$$

where ζ_R denotes the Riemann zeta function. The envelope derivatives are uniformly bounded as w decreases.

Old body–line cross terms also remain. Let S_f be the exact line-to-line scattering operator and $g_n = S_f^*(i\zeta\partial_t + i\zeta'/2)v_n$. Its delay kernel contains shifts kw. Since $e^{-in\Omega kw} = 1$, direct substitution gives $g_n = e^{-iE_n t}(E_n a_w + b_w)$, $E_n = \Delta + n\Omega$, with all derivatives of a_w, b_w uniformly bounded independently of n, Δ. The wrong KMS occupation of this unnormalized mode is bounded by

$$2(E_n^2\|a_w\|_2^2 + \|b_w\|_2^2)e^{-|E_n|/(2T)} + 2(2/|E_n|)^{2m}(E_n^2\|a_w^{(m)}\|_2^2 + \|b_w^{(m)}\|_2^2).$$

Sum this over all integers n and the four species to obtain G_T. The sum converges for $m \geq 2$. Its zero term tends to zero with Δ; its nonzero terms tend to zero with $\Omega \geq 2\Delta$. CAR and Cauchy–Schwarz bound the difference of the actual cross stresses by $4\sqrt{4 + M_E}\sqrt{G_T}$, without a product-state premise. The incoming-only term cancels because body-only interventions leave the same incoming marginal unchanged. The full lower bound is therefore

$$\begin{aligned}\Delta J_0(D - 16d - 2M_E) - 2(\Delta + \Omega)R_2(M_Q + 4\sqrt{M_Q}) \\ - 4\sqrt{4 + M_E}\sqrt{G_T}.\end{aligned} \tag{72}$$

Choose Γ large, then Δ large, then w small. The reset bounds and the displayed convergent tails make the error less than $\Delta J_0 D/2$. These choices remain compatible with the pump positivity condition (64). □

The energy difference is paid by preparing and maintaining different physical axes at a fixed gap. It is emitted by the reset itself, not by an additional program-tagged source. Conditional histories can have different incoming backgrounds; the causal result makes no claim that those observed conditional means obey the same formula.

D A spatial work clock for the retained thermal body

The waveforms above can be generated by a physical clock field over a specified finite horizon. Its source depends on the retained quantum state, including the body–line correlations created by earlier gates. For this construction supply a static material frame (U, V) and reflecting endpoints. Let

$$S_\chi = \frac{I}{2}\int \sqrt{-g}[(U\cdot\nabla\chi)^2 - C(V\cdot\nabla\chi)^2], \qquad 0 < C < 1, \tag{73}$$

with Neumann data, initially $\chi = 0$, $\chi_t = 1$, $\chi_x = 0$. Gate profiles are smooth functions of χ and the material coordinate. Use flat guard intervals between a zero-gap, fixed real quartet quadrature and the quadratic gap, pair and mixer gates. The actual gap-lowering and restoration work, offsets and finite normal-order terms are included in the Hamiltonian. Internally mixed components have a common positive free speed v_0.

Proposition 35 (Retained-state field–clock evolution)**.** *Fix a finite list of such windows and a finite horizon H. Prepare the original mixed four-mode target and its purifier, the empty-zero body reference and an actual finite-volume Gibbs reservoir; retain every subsequent state*

and reference. Assume smooth stress and kernel remainders, sufficiently many finite energy moments, and the strict quartet inequalities (77) *below. Choose the reservoir long enough to prevent a return before H. Then sufficiently large finite physical inertia I_H gives a timelike work clock and the same sequence of gate types throughout H. Its force is that of the actual evolved full-CAR state. The quadratic windows and their delayed matrix boundary close at the same derivative order. The quartet window admits a reflected symmetrizer at that order. In each specified lower source norm the field and the paid clock-work source have a comparison error $O_H(I^{-1/2})$; a quadratic window with identical incoming data has $O_H(I^{-1})$ error. These estimates retain the clock's $O(I)$ baseline stress and its finite work contribution.*

Proof. During a quadratic window the Heisenberg multiplet obeys $(\partial_t + \sigma v_0 \partial_x)\Psi = -iF\Psi$, where $F = F(\chi, a)$ is Hermitian and $\sigma = \pm 1$ is chirality. Pair gates use constrained Majorana components; their doubling adds no physical outlets. Subtract the fixed symmetric spatial singularity $-i\sigma\kappa_J/(x-y)$, $\kappa_J = 1/(2\pi)$, from $\Gamma(x,y) = \langle \Psi^\dagger(y)\Psi(x)\rangle$. Let $J = \Gamma_{\rm reg}(x,x)$ and $K = -i\sigma(\partial_x - \partial_y)\Gamma_{\rm reg}/2|_{y=x}$. Point splitting gives the full matrix equations

$$\begin{aligned} J_t + \sigma v_0 J_x &= -i[F, J] - \sigma\kappa_J F_x, \\ K_t + \sigma v_0 K_x &= -i[F, K] - \frac{\sigma}{2}\{F_x, J\}, \qquad F_x = F_\chi \chi_x + F_a a_x. \end{aligned} \tag{74}$$

The force in (73) is $-\sum_\sigma \langle F_\chi, J_\sigma\rangle - \partial_\chi E_o$. For complex components the contraction is a trace; for constrained Majoranas it is half the trace. Finite sea shifts are retained: the periodic empty-zero convention has $[L_n^{\rm sea}, J_m^{\rm sea}] = -mJ_{n+m}^{\rm sea} - (n/2)\delta_{n+m,0}$. Passing to canonical currents uses $J_n = J_n^{\rm sea} - \delta_{n0}/2$ and $L_n = L_n^{\rm sea} + \delta_{n0}/12$, with the corresponding fixed Hamiltonian subtraction. It leaves the physical zero occupation unchanged.

For a mixing arc of length $\ell < w$, set $d_a = \ell/v_0$, $d_{\rm out} = (w-\ell)/v_0$ and $d = w/v_0$. The arc Wilson matrix $G(u)$ maps entry time u to exit $u + d_a$. With its body/line blocks G_{BB}, G_{BF} and the actual outside phase Q, the incoming body satisfies

$$b(u) = A(u)b(u-d) + B(u)f(u-d), \quad A(u) = QG_{BB}(u-d), \quad B(u) = QG_{BF}(u-d). \tag{75}$$

Finite iteration expresses its cross kernel as

$$\Gamma_{BF}(u,v) = R_M \Gamma_{B_0F}(u - Md, v) + \sum_{j=0}^{M-1} R_j B(u - jd)\Gamma_{FF}(u - (j+1)d, v),$$

where $R_j = A(u)\cdots A(u-(j-1)d)$ and $R_0 = 1$. These are kernels of the original joint state. At $v = u$ every reservoir separation is at least d; all clock-dependent incoming data precede u by at least $d_{\rm out} > 0$.

The incoming cross stress requires covariant relative time differentiation. For constant exterior connections $F_{0,B}, F_{0,F}$,

$$K_{BF} = \frac{i}{2v_0}(\partial_u - \partial_v)\Gamma_{BF}|_{v=u} - \frac{F_{0,B}\Gamma_{BF} + \Gamma_{BF}F_{0,F}}{2v_0}.$$

For the full right-moving transfer put $S = (iG'G^\dagger + GF_{0,\rm in}G^\dagger - F_{0,\rm out})/v_0$. The complete spatial point-split conversion is

$$\begin{aligned} J_{\rm out} &= GJ_{\rm in}G^\dagger + \kappa_J S, \\ K_{\rm out} &= GK_{\rm in}G^\dagger + \tfrac{1}{2}\{S, GJ_{\rm in}G^\dagger\} + \tfrac{\kappa_J}{2}S^2. \end{aligned} \tag{76}$$

Distinct body/line finite-circle backgrounds are added before and subtracted after this conversion. Differentiating unitarity twice in the symmetric split cancels the anti-Hermitian second derivative, including noncommuting connection terms. A scalar total stress would lose the BF data in this equation.

Differentiate the arc Wilson matrix by Duhamel's formula. Its $(m+1)$st entry-time derivative needs only derivatives through m of χ_t, a first-order clock variable. The initial two-time kernel remainders and their required derivatives are bounded away from the declared diagonal singularity. Finite history products therefore give a locally Lipschitz C^m boundary functional in (χ_t, χ_x, J, K), $m \geq 3$. On time steps shorter than d_{out} its incoming data are already fixed. Characteristic iteration of (74) and the acoustic-clock wave equation uses these same jets and yields the quadratic-window estimate. The constants depend on H/d and the initial norms.

For a quartet window write $v_s = v_0 + sa(\chi)$, $g_s = sa_\chi$, $s = \pm$, and $\gamma_s = 1/(48\pi)$. The actual chiral stress satisfies $T_{\sigma s,t} + \sigma v_s T_{\sigma s,x} + 2\sigma v_{s,x} T_{\sigma s} = \sigma \gamma_s v_{s,xxx}$. Use $Y_{\sigma s} = (1-\partial_x^2)^{-1} T_{\sigma s}$ and $Z_{\sigma s} = \partial_x Y_{\sigma s}$, retaining their means. For $r = \chi_t$, $p = \chi_x$, the principal matrix has $A_{rp} = -C$, $A_{rZ} = -g_s/I$, $A_{pr} = -1$, $A_{Zp} = \sigma\gamma_s g_s$, $A_{ZZ} = \sigma v_s$. Its symmetrizer has $H_{rr} = I$, $H_{pp} = IC$, $H_{pZ} = g_s$, $H_{ZZ} = v_s/\gamma_s$. Direct Schur elimination gives

$$IC > 2\gamma_s a_\chi^2 \sum_s v_s^{-1}, \qquad I(1-C) > 2\gamma_s a_\chi^2 \sum_s \frac{v_s}{1-v_s^2}. \tag{77}$$

Thus H and $H \pm HA$ are positive. The exact remainder follows by commuting $1-\partial_x^2$ through the Ward equation; its terms with $v_{s,xx}Z$ gain one derivative under $\partial_x(1-\partial_x^2)^{-1}$. They are locally Lipschitz at the same Sobolev order. At a reflected endpoint $p = 0$, $Y_{Rs} = Y_{Ls}$, $Z_{Rs} = -Z_{Ls}$, so the energy flux vanishes. Smooth higher classical energies recover the desired lower-norm actual stress. Increasing finite I preserves the flat guard intervals and a timelike clock over the fixed finite list.

For the solved smooth profiles, the full-tower propagator carries the actual state through each seam. Its distributions satisfy these same linear Ward and kernel equations; uniqueness identifies the mean force with that retained state. Finitely higher initial graph and classical norms can be selected for the finite list. In a gate controlled by a conserved record projector use unnormalized weighted kernels; each c-number anomaly carries that projector's actual weight. The controlled unitary also retains off-diagonal record coherence. Register updates occupy their separately paid slots.

Finally the actual clock work is

$$\frac{d}{dt}\frac{I}{2}\int (r^2 + Cp^2) = -\int r\,[\sum_\sigma \langle F_\chi, J_\sigma\rangle + \partial_\chi E_o],$$

with the quartet force replacing the bracket during its slot. Write $\chi = t + \eta_I/I$ and solve $\eta_{0,tt} - C\eta_{0,xx} = -\mathrm{force}_{\mathrm{ref}}$ with the same corrected initial data. The source estimate controls $\eta_I - \eta_0$; the finite retarded work term and the $O(I^{-1})$ quadratic clock remainder remain in the comparison. No growing clock baseline multiplies an uncontrolled state error. □

Corollary 36 (Full joint accuracy of the finite schedule)**.** *Use the original charge-glued field space, its finite reservoir and all retained quantum registers and references in Proposition 35.*

In fixed energy units put $G_{\mathrm{en}} = 1 + \sum_{\mathrm{rails}} v_0 P_{\mathrm{rail}}$, the positive physical excitation energy, including holes and every spectator. The finite zero-mode multiplicity is retained. Suppose the joint initial state has $M_1^2 = \mathrm{Tr}(\rho_0 G_{\mathrm{en}}^2) < \infty$, with the finitely higher smooth input norms required for the schedule. Compare its solved clock history with the prescribed history on the same field space and in the same initial state. If

$$a_I(t) = \|(H_I(t) - H_0(t))G_{\mathrm{en}}^{-1}\|, \qquad b(t) = \|[G_{\mathrm{en}}, H_0(t)]G_{\mathrm{en}}^{-1}\|,$$

then, for every $t \le H$, the half trace distance of their full joint quantum outputs is bounded by

$$e_H = M_1 \int_0^H a_I(s) \exp\left(\int_0^s b(r)\, dr\right) ds = O_H(I^{-1/2}). \tag{78}$$

At a reset checkpoint whose prescribed-history whole-body error is d_{ring} in (58), the actual output satisfies

$$\tfrac{1}{2}\|\rho^I_{\mathrm{body},R} - |\mathrm{sea}\rangle\langle\mathrm{sea}| \otimes \rho^I_R\|_1 \le d_{\mathrm{ring}} + 2e_H. \tag{79}$$

Here R is the actual outside system, including every old reference. The same joint bound applies after preparing the original mixed receiver with its current purifier and before its next specified readout. It is an energy-constrained state estimate, not an unconstrained diamond-norm assertion.

Proof. The local quartet is a stress smearing in its original sector representation. The positive-energy bound [3, Sections 2.2–2.3]

$$\|L_n\psi\| \le \sqrt{1 + c/12}(1 + |n|^{3/2})\|(1 + L_0)\psi\|$$

and Fourier Cauchy–Schwarz control a profile difference by its H^m norm, $m > 2$. The energy commutator needs one further derivative. For a quadratic current, pair or mixer, separate the same-sea number part from the pair part. The first is bounded relative to excitation number and hence to G_{en} on the fixed finite hardware. A smooth compact-arc profile gives pair matrix entries bounded by $C_s(1 + E_n + E_m)^{-s}$. A sum-frequency shell has $O(q)$ entries: absolute summation requires $s > 2$, and the energy commutator requires $s > 3$. Three and four integrable spatial derivatives, respectively, are safe integer choices. Different circumferences enter through their physical energies and finite spacing constants; an unweighted sum of L_0's would not supply this mixer commutator. The estimates act across the original CAR sectors, including record coherences and their scalar phases. They use no Gaussian replacement at a seam.

The smooth propagators preserve the common energy core. For a purification Ψ_0 of the actual initial state, the commutator estimate gives $\|G_{\mathrm{en}}U_0(s)\Psi_0\| \le M_1 \exp(\int_0^s b)$. Duhamel's formula then bounds $\|(U_I(t) - U_0(t))\Psi_0\|$ by (78); every reference carries the identity. The clock comparison at the chosen lower coefficient norm gives $\int_0^H a_I \le C_H I^{-1/2}$, after supplying the required higher norms. Its existence interval already covers the entire schedule. Triangle inequality with the prescribed reset costs e_H; replacing its outside marginal by the actual one costs another e_H. Re-preparation and the same subsequent instrument contract this error. □

The delay in reset bounds is the round-trip time d, whereas w is spatial length. For example, $v_0 = .5$, $w = 10^{-8}$, $\Delta = 10^7$, $T = \tau = 1$ and $\Gamma = 32$ admit a smooth flat reset pulse with $D_0 < 9.9 \times 10^8$ and $r_d < 4 \times 10^{-5}$. The four-mode joint reset bound is 0.026503. Finite additional clock accuracy is obtained by increasing I_H; this argument does not provide its numerical value. Restoring the gap preserves the original X/Z work difference $\Delta(1-2q)(1-2\epsilon)$. For a constant arc background $F_0 = \operatorname{diag}(\Delta, 0)$ the actual local connection

$$F = F_0 + v_0 h(a) e^{-iF_0 a/v_0} W(\chi - a/v_0) e^{iF_0 a/v_0}, \qquad \int h \, da = 1,$$

gives the full circuit phase $e^{-i\Delta d}$, retaining its spatial force and emitted carrier energy.

The evolving work clock still has a supplied frame/endpoints; traction and geometric reaction need their own equations. Equal mixer speeds, smooth inputs and separated quartet/current slots are required. Horizon-dependent inertia and reservoir length leave fixed-capacity sharp exhaust open.

D.1 A local gate on a massive body

A spatial body need not have the massless ring's energy graph. Pay instead for a finite physical circle with a mass-zero central interval of length ℓ and a positive smooth Dirac mass throughout the rest of its return arc. Its four thermal outlet fields remain separate, with the original charges and capacity. Both mass layers, the exterior polarization, support pressure and initial preparation belong to the apparatus. This layout replaces a compact barrier followed by a long gapless body exterior; it does not give that old layout a gap by assumption.

Proposition 37 (A guarded local gate with full physical energy control)**.** *Let $h_M = -iv\sigma_z\partial_x + M(x)\sigma_x$ on this circle. There is a nonempty smooth mass-profile class with a simple lowest positive mode E_0, no zero mode and next positive energy $E_1 > E_0$. Set $H_M = d\Gamma(|h_M|)$, with particles and holes above the actual filled mass sea, and $A = 1 + H_M$. Let $h \geq 0$ be smooth, supported strictly inside the mass-zero region. On the original charge-glued space define the local vertex*

$$Q_+(h) = \int h(x)[R_1^\dagger R_2^\dagger R_3^\dagger R_4^\dagger + L_1^\dagger L_2^\dagger L_3^\dagger L_4^\dagger]dx, \qquad Q = Q_+ + Q_+^\dagger.$$

All modes and old references are retained. For each fixed graph order s, sufficiently weak smooth resonant drives

$$H_{\mathrm{lab}}(t) = H_M + \lambda b(\lambda t)[e^{-4iE_0 t} Q_+(h) + e^{4iE_0 t} Q_+(h)^\dagger] \tag{80}$$

implement a prescribed full logical angle on the four-ground-mode code P_0, in physical time $T = U/\lambda$, with

$$\|A^s(U_{\mathrm{lab}}(T) - U_{\mathrm{target,lab}}(T))P_0\| \leq C_s\lambda. \tag{81}$$

The target includes its calibrated laboratory phase. The error is uniform in external old references, not an unrestricted-energy diamond bound. Smooth physical mass-force and stress tests satisfy $\|A^{-4}S_f A^{-4}\| < \infty$, so weighted state errors also control their actual renormalized expectations at this fixed mass.

Proof. The original orthogonal boson rotation and re-fermionization give, for each chirality, the operator identities

$$Q_\sigma(h) = \pi^{-1}(T_{\sigma,1}(h) - T_{\sigma,2}(h)), \qquad T_{\sigma,1} + T_{\sigma,2} + \sum_{a=1}^{3} T_{\sigma,a} = \sum_{i=1}^{4} T_{\sigma,\mathrm{Dirac},i}.$$

The individual rotated stresses need not be quadratic in the massive species. Their total on the right is quadratic. The commuting sector stresses have positive shifted forms by the chiral QEI [12, Theorem 4.1 and Remarks 2–3]. Their joint spectral calculus therefore bounds $\|Q(h)\psi\|$ by $C\|(K_h + C_h)\psi\|$, where K_h is the actual original localized kinetic stress. Global charge gluing is retained; no independently prepared sector vacua are supplied.

In the physical mass polarization its one-particle operator is

$$D_h = -iv\sigma_z(h\partial_x + h'/2) = hh_M - iv\sigma_z h'/2,$$

since $hM = 0$. For a number block b and $\omega = |h_M|$, $b^2 \le C^2\omega^2$ and hence $|b| \le C\omega$. On an n-particle tensor space, different-particle factors commute, so

$$(\sum_i b_i)^2 \le \sum_i b_i^2 + 2\sum_{i<j} |b_i||b_j| \le C^2(\sum_i \omega_i)^2.$$

This gives a uniform Fock bound without a spurious $\sqrt{n}$. The number-block closure follows from the commutator criterion with $1 + \omega^2$ on the smooth elliptic core. On the guard all mass jets vanish. The complete symbols of the spectral projections there agree with the massless chiral projections; consequently $P_+ D_h P_-$ is smoothing. Its actual mass-eigenbasis coefficients decay faster than every fixed power of $E_n + E_m$. Number and pair estimates give QA^{-1} bounded. For signed derivatives use a wider guarded positive cutoff, not a bound on h'/h. Local translation gives $\mathrm{ad}^j_{H_M} Q(h) = (-iv)^j Q_R(h^{(j)}) + (iv)^j Q_L(h^{(j)})$. Thus every fixed required energy commutator is relatively bounded and the smooth weak-drive propagator preserves the corresponding graphs over its entire gate time.

The physical spectral class can be constructed, rather than postulated. For a step reference with mass m on the exterior arc of length d, the subthreshold parity equations are

$$\tan\frac{E\ell}{2v} = \sqrt{\frac{m-E}{m+E}}\tanh\frac{d\sqrt{m^2-E^2}}{2v} \quad (+),$$

with the negative reciprocal square-root on the right for parity $(-)$. If $m > 3\pi v/(2\ell)$, the first positive root is simple, $0 < E_0 < \pi v/(2\ell)$, while $E_1 > \pi v/\ell$. At zero the transfer is $\exp((\int M/v)\sigma_y)$ and has no unit eigenvalue. Smooth the two steps just outside the body. Transfer Duhamel bounds the perturbation on fixed complex contours by $C\|M_{\mathrm{smooth}} - M_{\mathrm{step}}\|_1$; choosing it below the positive minimum of the explicit step characteristic function preserves these root counts and a strict gap. This gives a real smooth class, without treating a sampled eigengrid as its proof. In the body its normalized ground mode has $R = e^{iE_0x/v}/\sqrt{2Z}$ and $L = e^{-iE_0x/v}/\sqrt{2Z}$. For $\operatorname{supp} h \subset (-\ell/8, \ell/8)$,

$$g_4 = \frac{1}{2Z^2}\int h\cos(4E_0x/v)dx > 0, \qquad g_2 = \frac{1}{Z}\int h\cos(2E_0x/v)dx > 0.$$

Let N be signed flavour number, particles minus holes. It is not excitation number or total electric charge. Then $K_0 = H_M - E_0 N \geq 0$ has the sixteen-dimensional kernel P_0 and gap $\delta = \min(E_1 - E_0, 2E_0) > 0$; its graph is equivalent to A. Because $[N, Q_+] = 4Q_+$, $e^{-iE_0 tN}$ transforms (80) exactly to $K_0 + \lambda b(\lambda t)Q$. This is a proof coordinate, not a physical global-number pulse. The compression is $P_0 Q P_0 = g_4 C_4^\dagger + \overline{g}_4 C_4$; all off-code terms remain in the laboratory Hamiltonian.

On $|z| = \delta/2$ the weighted resolvent norms $\|A^j Q(K_0 - z)^{-1} A^{-j}\|$ are finite. Choose $\lambda\|b\|_\infty$ within their Neumann tube and the physical relative-bound margin. The contour projection and its Kato transport give an effective code matrix $aP_0QP_0 + O_s(a^2)$, $a = \lambda b$. Its integrated remainder is $O_s(\lambda)$ because $\int_0^T a^2 dt = \lambda \int_0^U b^2 du$. The transport residual costs $\int_0^T |\dot{a}| dt = \lambda \,\mathrm{TV}(b)$. Energy propagation costs $\exp(C_s \int |b| du)$, independent of λ. Choosing $\int b du = \theta/g_4$ proves (81). Pair resonance at $2E_0$ and guarded currents supply the paired-code rotations and phase. For $\theta = \arcsin\sqrt{\epsilon}$ the vacuum preparation is the original mixed A/R target, at final energy $4E_0\epsilon$ up to the weighted error, supplied by the pump.

Finally a smeared physical mass bilinear has order zero and a metric stress order one. A safe pair coefficient bound is $C_f(1 + E_n + E_m)^d$, $d \leq 1$. Weighting both sides by A^{-4} bounds each creation term by $(1 + E_n + E_m)^{-4}$. One-dimensional energy-shell counting has $O(q)$ pairs, so the absolute sum converges. Number terms obey the previous estimate. With one fixed Hadamard subtraction scheme, for $d_4 = \frac{1}{2}\|A^4(\rho - \sigma)A^4\|_1$ this yields $|\operatorname{Tr} S_f(\rho - \sigma)| \leq 2\|A^{-4} S_f A^{-4}\| d_4$. Counterterm changes between backgrounds are not cancelled by this fixed-background comparison. □

This gate acts on the full massive carrier, with the actual pump phase, work and wall source. It does not use a hardware projector onto four smeared modes. Its source certificate is for smooth tests at a fixed mass, not pointwise source jets on moving walls. The massive carrier's outlet mixer and whole-sea renewal must be computed on its own spectral generator; the earlier massless ring's reset coefficients cannot be reused. Nor are its ground modes already the massless short-ring storage modes in Proposition 32. Physical transfer, moving support and a reused work clock must join these layouts before they form one sustained device.

E A finite conserved control core

A finite prescribed circuit, energy supply and thermal ports can share a time-independent Hamiltonian. We combine energy-ladder control [20] with a finite clock [22], using complete energy sectors and an exactly solvable history clock. Register, gate permissions and initial resources are supplied inputs.

E.1 A guarded battery for the whole circuit

Let $H_D = \delta \sum_a q_a |a\rangle\langle a|$, with integers $0 \leq q_a \leq Q$, and let $H_B = \delta \sum_{n=0}^K n|n\rangle\langle n|$, $K \geq 2Q$. The register D includes its program, retained system and all thermal ports; an inert reference

can be appended. Define

$$V(U)|a, E - q_a\rangle = \sum_b U_{ba}|b, E - q_b\rangle, \qquad Q \le E \le K, \tag{82}$$

for any unitary U, and take the identity on incomplete energy sectors.

Proposition 38 (Finite energy-conserving circuit lift)**.** *The operator $V(U)$ is unitary, commutes with $H_0 = H_D + H_B$, and $V(U)V(W) = V(UW)$. Put $K = 2Q + L - 1$ and prepare*

$$|\beta_L\rangle = \sqrt{\frac{2}{L+1}} \sum_{j=1}^{L} \sin\frac{\pi j}{L+1}|Q + j - 1\rangle, \qquad e_B = 2\sin\frac{\pi}{2(L+1)}.$$

For any input of D entangled with a reference and initially independent of this battery, the joint half trace distance between the lifted circuit $V(U_m)\cdots V(U_1)$ and $U_m \cdots U_1 \otimes I_B$ is at most $\min\{1, 2Qe_B\}$. The comparison retains the battery and all correlations.

Proof. Every complete energy sector carries a copy of the entire data space; the others carry the identity. This proves unitarity, conservation and composition. Extend the battery vector by zero to a bilateral ladder with shift S. Its adjacent overlap is $\cos[\pi/(L+1)]$, so $\|(S^k - I)\beta_L\| \le |k|e_B$. With $D_q = \sum_a |a\rangle\langle a| \otimes S^{-q_a}$, $V(U) = D_q(U \otimes I)D_q^*$ on the guarded inputs. Each dressing difference has vector norm at most Qe_B, even with a retained reference. Their sum bounds the joint state distance after purification. The guards make this bilateral calculation agree with the finite-sector action; no shift wraps through an endpoint. □

The bound concerns the whole compiled circuit at a fixed register energy range. Fresh thermal units in a growing protocol enlarge Q. Battery coherence, mean energy and preparation remain resources; the battery is not an exact catalyst.

E.2 Autonomous timing and a usable output window

Let $V_j = V(U_j)$ and add clock states $|0\rangle, \ldots, |m\rangle$. For $J > 0$ and $\Lambda = Jm$, set

$$H = H_0 + \Lambda I + \sum_{j=1}^{m} J\sqrt{j(m-j+1)}\big(|j\rangle\langle j-1| \otimes V_j + \text{h.c.}\big). \tag{83}$$

The clock-dependent interaction is part of the apparatus energy.

Proposition 39 (Positive autonomous execution)**.** *The Hamiltonian is nonnegative and commutes with H_0. From clock state $|0\rangle$, at $T = \pi/(2J)$ it gives*

$$e^{-iHT}(|0\rangle \otimes |\psi\rangle) = e^{-i\Lambda T}(-i)^m|m\rangle \otimes e^{-iH_0T}V_m \cdots V_1|\psi\rangle.$$

If only the first $r \le m$ edges perform gates and the rest are identities, then throughout $[T/2, 3T/2]$ the data's half trace distance from the completed lifted circuit, including its free evolution, is at most

$$P_{m,r} = \Pr\{\mathrm{Bin}(m, 1/2) < r\}. \tag{84}$$

The joint data–clock distance to the completed data and the normalized completed-tail clock state is at most $\sqrt{P_{m,r}}$.

Proof. Put $W_j = V_j \cdots V_1$, $W_0 = I$, and $\mathcal{W} = \sum_j |j\rangle\langle j| \otimes W_j$. Conjugation by $\mathcal{W}^*$ reduces the hopping term to the spin-$m/2$ chain $2JJ_x$, commuting with H_0. Its spectrum lies in $[-Jm, Jm]$; the shifted block lies in $[0, 2\Lambda]$. The spin rotation transfers 0 to m at T, as in perfect-transfer chains [21]. Its position law is $\mathrm{Bin}(m, \sin^2 Jt)$. Every prefix $j \geq r$ has the same completed data. Tracing the clock leaves at most the incomplete probability as error. Projection onto the completed tail has squared overlap one minus this probability, giving the square-root joint bound. On the stated window $\sin^2 Jt \geq 1/2$. □

The joint target uses the normalized tail, not the unchanged full clock wavefunction; the latter comparison instead has bound $2\sqrt{P_{m,r}}$. For $r = 10$, $m = 64$ and $\Lambda = .4$, $P_{m,r} \approx 1.77111 \times 10^{-9}$ and $T \approx 251.327$ in the declared energy units. The clock has 65 states. This finite window is not permanent halting: the closed clock eventually returns and reverses its gates. Its initialization, idle states and any output receiver remain charged resources.

E.3 Two cycles through the same thermal contact

The core has seven qubits: detector S, retained body Q, contact F, output cells E_1, E_2 and dephasing cells D_1, D_2. The body and dephasing cells have zero bare gap. The other four have gap δ, so the register has maximum charge 4. Prepare S, F, E_1, E_2 independently in $\tau = \mathrm{diag}(1-\epsilon, \epsilon)$, $\beta\delta = \ln[(1-\epsilon)/\epsilon]$, and each D_i in its zero-gap Gibbs state $I/2$. The body may be entangled with a retained reference and is never re-prepared.

Cycle i consists, in order, of

$$e^{-i\theta_i X_S P_Q}, \quad P_Q = |1\rangle\langle 1|_Q, \quad \mathrm{CZ}_{S,D_i}, \quad \mathrm{CX}_{S,Q}, \quad \mathrm{SWAP}_{S,F}, \quad \mathrm{SWAP}_{F,E_i}\,.$$

The dephased detector directly controls the body operation. Each gate is lifted and placed on an edge of Eq. (83); no external switching times enter. The first swap transfers the old detector and all its correlations to F, putting an independent Gibbs state in S. The second moves the old state to E_i and restores F from the unused Gibbs cell. Both conserve bare energy. Replacement is exact against every retained reference, including the battery, and returns a thermal state rather than a pure blank.

The first recorded bit has probability ϵ when the body is $|0\rangle$ and $\epsilon + (1-2\epsilon)\sin^2\theta_1$ when it is $|1\rangle$. This is a genuine body reading. The command acts on the already retained body and can damage its initial coherence; it is not a coherence-preservation claim. For $\epsilon = .1$, $\delta = .1$, $\theta_1 = \pi/3$ and $\theta_2 = \pi/2$, with the body initially half of a Bell pair, output populations are $.4, .26$. Net heat $Q_E = .046$ equals the battery's mean energy loss. The joint output obeys

$$\beta Q_E = S(\rho_{E_1E_2}) - 2S(\tau) + D(\rho_{E_1E_2} \| \tau \otimes \tau). \tag{85}$$

This follows from $-\ln\tau = \beta H_E + \ln Z$. Other ports, the body, reference, clock and battery remain in the full entropy account.

At $L = 31$ the battery has 39 levels and mean initial energy 1.9. The two sensing rotations give a joint error bound $(\sin\theta_1 + \sin\theta_2)e_B = .183123$; the full purification gives the smaller upper bound .070347. For $\Lambda = .4$, the complete Hamiltonian has cutoff 5.0 and initial mean 2.34. At $L = 127$, the purification upper bound is .017610, mean energy 7.14 and cutoff 14.6. These are executions retaining the complete correlated state.

The cells hold a final energy .066, including .02 initially present; net heat .046 is not automatically emitted radiation. Spatial transport, release recoil and a stationary thermal outlet are further operations. HAFF D gives a material-support branch compatible with the retained core's mean energy [23]. It changes the support's dilaton coupling; the spin cells are not identified with the preceding continuum.

F A local outlet with an autonomous clock

Heat stored in a cell can leave through a local Hamiltonian. Let the output cell be site 0 of a semi-infinite fermion chain,

$$H_{\mathrm{lead}} = \omega \sum_{j\geq 0} n_j + g \sum_{j\geq 0} (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j), \qquad \omega > 2g > 0. \tag{86}$$

Finite open chains have a nonnegative many-body Hamiltonian and spatially nearest-neighbour coupling. Equivalently they are open spin chains with XY exchange under the Jordan–Wigner map; the reference-uniform cell channel below uses this tensor-product realization. The infinite formula denotes its local dynamics, not a finite total-energy thermal bath. Initially site 0 has population p and may be correlated with old references. The β-KMS bath on $j \geq 1$ is initially independent of that entire cell–reference state. Use as comparison the same bath and the site population $\epsilon = (1 + e^{\beta\omega})^{-1}$.

Proposition 40 (Spatial export and reduced thermalization)**.** *Write $w_j(t)$ for the one-particle propagator from site 0. The excess covariance and on-site energies are*

$$\Delta\Gamma(t) = (p - \epsilon)|w(t)\rangle\langle w(t)|, \qquad \Delta E_j = \omega(p - \epsilon)|w_j(t)|^2. \tag{87}$$

For the homogeneous chain the excess hopping energy is zero, and the energy beyond the cell is $\omega(p - \epsilon)(1 - |w_0|^2)$. On the semi-infinite chain,

$$\alpha(t) := w_0(t) = e^{-i\omega t} \frac{J_1(2gt)}{gt}. \tag{88}$$

Let $b(t)$ be the population produced at site 0 from the bath alone and $n_{\mathrm{base}}(t) = |\alpha|^2\epsilon + b(t)$. Against every old external reference, the reduced cell's half trace distance from its Gibbs replacement is at most

$$\min\{1,\ |\alpha|^2 + |\alpha|/2 + |n_{\mathrm{base}}(t) - \epsilon|\}. \tag{89}$$

The retained lead and its initial purification are excluded from that product replacement claim.

Proof. Quadratic fermion evolution sends a covariance to $\Gamma(t) = U(t)\Gamma(0)U(t)^*$, giving the rank-one difference. On a bipartite chain successive components of w have relative phase $-i$ after removing $e^{-i\omega t}$, so $\mathrm{Re}(w_j^* w_{j+1}) = 0$. The surface spectral density is semicircular; its Fourier transform gives Eq. (88). This is standard tight-binding propagation with non-Markovian returns [19, Sec. III]. The thermal bath is even in fermion parity. The cell channel therefore sends population p to $|\alpha|^2 p + b$ and off-diagonal entry ρ_{01} to $\alpha^*\rho_{01}$. For a state with reference blocks ρ_{ab}, the diagonal difference from replacement at population n_{base} is $|\alpha|^2[(1 - \epsilon)\rho_{11} - \epsilon\rho_{00}]$, of trace norm at most $|\alpha|^2$. Positivity gives $\|\rho_{01}\|_1 \leq 1/2$. Adding the change from n_{base} to ϵ proves the bound. □

The lead need not be exactly Gibbs for the bare cell gap. The bath band has width $4g$, and the derivative of the Fermi function is bounded by $\beta/4$, so its occupation differs from ϵ by at most $\beta g/2$ throughout that band. Narrowing g controls this mismatch but lengthens transport. Equation (88) decays as $t^{-3/2}$ with returns and zeros; energy export is not monotone. Finite leads eventually recur. For a chain with sites $0, \ldots, N$, the finite and infinite wavevectors agree through Taylor order N. For $N+1 > 2gt$, a norm bound on their difference is

$$\frac{2(2gt)^{N+1}}{(N+1)!}\left(1-\frac{2gt}{N+2}\right)^{-1}.$$

This bounds propagation, not a global distance between finite and infinite thermal states.

For the two output cells with populations $.4, .26$, use two identical leads with $\omega = .1$, $g = .01$, $\beta = \ln 9/.1$. Their total initial excess is .046. At the first surface-amplitude zero, $t = 191.5853$, this excess is outside the cells. A finite calculation with 256 bath sites per lead gives $n_{\text{base}} = .10142988$. At $t = 2000$, excess energy .04599817 is in the leads and .04422294 is beyond site 16. Each bath initially contains energy 2.64854457. These values are transport after circuit completion; they do not discard the output correlations or identify a lattice energy profile with a null CFT stress tensor.

Reduced thermalization also differs from the exact SWAP of Appendix E. A two-site passive exchange at its perfect-transfer time is an iSWAP. Acting on $|+\rangle\langle+| \otimes \operatorname{diag}(.9, .1)$, it gives the cell exactly that Gibbs marginal, but the joint half trace distance from the product of its two marginals is .18. The bath cannot be called fresh for later cell–bath encounters.

F.1 The outlet can remain on during computation

Let H_{hist} be Eq. (83), with only its first r edges active, and let $H_B \geq 0$ be a finite bath Hamiltonian. For a local outlet interaction V_o with $\|V_o\| \leq v$, take $H_{\text{tot}} = H_{\text{hist}} + H_B + V_o + vI$. It is nonnegative and time independent. The offset v is part of its mass and energy account.

Proposition 41 (Always-on outlet bound)**.** *Put $T = \pi/(2J)$, choose $0 < t_0 < T$, and define*

$$P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \leq r\}, \qquad A_r = J\sum_{j=1}^{r}\sqrt{j(m-j+1)}.$$

Compare the actual joint state at $t \in [t_0, T]$ with completed lifted data, including free evolution to t_0, and the free clock wavefunction at t_0, subsequently evolved with the outlet and identity clock edges. Keeping every reference, their half trace distance is at most

$$\min\{1,\ vt_0 + 2\sqrt{P_0} + 4A_r(t-t_0)\sqrt{P_0}\}. \tag{90}$$

Proof. Removing the always-on outlet only up to t_0 costs at most vt_0 by Duhamel's formula. Replacing all prefixes by the completed data costs at most $2\sqrt{P_0}$ in purification norm. After t_0, compare with the same outlet and free clock; only the first r history edges differ. The free-clock probability of positions $\leq r$ is at most P_0 through T. The norm of the differing edges on that state is at most $4A_r\sqrt{P_0}$. A second Duhamel bound gives the last term. No independence of the output cells or later fresh-bath composition is used. The index $\leq r$ includes the last active edge. □

For any finite transport window the bound can be made small by increasing the idle clock and its energy. For example $r = 10$, $m = 10^{10}$, $J = .001$, $\sin^2 Jt_0 = 128/m$ and two outlets with $v = .02$ give $t_0 = .1131371$, $T = 1570.7963$ and the sufficient joint bound .002262742. This uses $10^{10} + 1$ clock states and clock mean energy 10^7; it proves finite compatibility, not an efficient clock. A direct smaller Hamiltonian with 17 clock states, five battery levels and outlet norm 10^{-5} gives joint purification distance .00015225 below its certificate .00025067. Thus spatial export need not be externally switched, but a fixed-size indefinitely operating exhaust is not obtained.

G Thermal sources and their preparation

A conformal fermion defect gives a local relativistic alternative to the lattice outlet. The real-time defect construction and its thermal current are established results [17, Sections 2, 3 and 5]. Take two independent copies of the Majorana construction, so each port is a complex chiral fermion with $c = 1$. In the folded description, a constant contact angle acts on the two fields at the same contact time:

$$\begin{pmatrix} \psi_1^{\mathrm{out}} \\ \psi_2^{\mathrm{out}} \end{pmatrix} = S_\theta \begin{pmatrix} \psi_1^{\mathrm{in}} \\ \psi_2^{\mathrm{in}} \end{pmatrix}, \qquad S_\theta = \begin{pmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{pmatrix}, \quad \chi = \sin^2\theta. \tag{91}$$

The inputs are jointly independent, zero-charge quasifree thermal streams in one specified time coordinate, at $T_H < T_o$. This assumes a prepared reservoir, rather than an already prepared outgoing field.

Proposition 42 (Thermal transfer and its retained correlations)**.** *For $\omega > 0$ put $n_T = (1 + e^{\omega/T})^{-1}$. The source-port occupation and transferred energy current are*

$$n_\chi = (1-\chi) n_{T_H} + \chi n_{T_o}, \qquad q_\chi = \frac{\pi\chi}{12}(T_o^2 - T_H^2). \tag{92}$$

Its cross covariance with the return port has modulus $\sqrt{\chi(1-\chi)}(n_{T_o} - n_{T_H})$. For $0 < \chi < 1$ the source is not a KMS state at any single temperature; at $\chi = 1$ it is exactly the supplied hot state. Defining its stationary marginal entropy-current excess, with $h(x) = -x\ln x - (1-x)\ln(1-x)$, by

$$s_\chi = \frac{1}{\pi}\int_0^\infty [h(n_\chi) - h(n_{T_H})]\, d\omega,$$

one has

$$q_\chi \geq T_H s_\chi + \frac{3 s_\chi^2}{\pi}, \tag{93}$$

strictly for $0 < \chi < 1$.

Proof. Rotate the full input covariance with S_θ. The factor $1/\pi$ in the integrals includes particles and holes of a complex chiral fermion; $\pi^{-1}\int \omega n_T d\omega = \pi T^2/12$. At high frequency $n_\chi \sim \chi e^{-\omega/T_o}$ for $\chi > 0$, whereas a single-temperature Fermi occupation has tail coefficient one. For fixed energy the Gibbs variational principle maximizes entropy at $T_{\mathrm{eff}} = \sqrt{T_H^2 + 12 q_\chi/\pi}$. Hence $s_\chi \leq \pi(T_{\mathrm{eff}} - T_H)/6$, with equality only for that thermal state. Eliminating T_{eff} gives the bound and its strictness. □

The return occupation is $\chi n_{T_H} + (1-\chi) n_{T_o}$. Both outputs must be kept when accounting for entropy and correlations. This is not thermal replacement of an arbitrary qubit against every retained reference. If the inputs have a cross covariance, even their thermal marginals do not imply Eq. (92). A finite matched-frequency check uses the positive Hamiltonian $H = \omega(n_1 + n_2) + g(c_1^\dagger c_2 + c_2^\dagger c_1)$, $0 < g < \omega$. It commutes with the free energy and transfers equal energy out of the reservoir and into the source. This mode check does not establish a continuum limit for a microscopic moving contact.

Changing the contact has a cost in the same output fields. For a smooth prescribed $\theta(t)$ the full unequal-time covariance is $C_{\text{out}}(t,t') = S(t)\,\mathrm{diag}(C_H, C_o)(t-t')S(t')^\dagger$. The short-distance kernel is $C_T(s) = iT/[2\sinh(\pi T s)]$. At $t \pm s/2$, its common vacuum term is multiplied by $\cos[\theta(t+s/2) - \theta(t-s/2)]$. Point splitting therefore gives an added current $\dot\theta^2/(4\pi)$ in each output, so the total work supplied to the fields is

$$W_{\text{pump}} = \frac{1}{2\pi}\int \dot\theta^2 dt \geq \frac{(\Delta\theta)^2}{2\pi\Delta t}. \tag{94}$$

This agrees with the energy-shift dissipation form in adiabatic scattering [18, Eq. (8)]; here point splitting derives it exactly for the specified energy-independent contact. The inequality is Cauchy–Schwarz. A cosine ramp gives $W_{\text{pump}} = \pi(\Delta\theta)^2/(16\Delta t)$. Both ports' pump energy is charged; it is separate from the static net thermal transfer. A small angle error alone is no source certificate: $\theta = \theta_0 + \alpha\sin\Omega t$ has mean total pump current $\alpha^2\Omega^2/(4\pi)$. Instantaneous thermal entropy formulas do not apply during this switch. The controller that supplies this work and the contact's material stress still require their own realization. The input preparation has the following finite-window alternative.

G.1 Thermal correlations from a finite-energy preparation

Fix a bounded interval I of length L and $T > 0$. The geometric KMS construction is locally a vacuum pullback by the exponential map [10, Section 2.8, Proposition 2.4 and Theorem 2.5]. Its compact completion permits a finite global preparation. Use the state convention

$$C_\rho(x,y) = \frac{i\sqrt{\rho'(x)\rho'(y)}}{2\pi[\rho(x) - \rho(y) - i0]}, \qquad \langle T(x)\rangle_\rho = -\frac{c}{24\pi}\{\rho, x\}.$$

Here $c = 1$ for each complex-fermion stream, and the inverse diffeomorphism implementer acts on the vacuum.

Proposition 43 (A thermal restriction with finite global energy)**.** *There is a smooth increasing $\rho : \mathbb{R} \to \mathbb{R}$, equal to the identity outside a compact interval, whose vacuum-orbit state has the complete zero-charge thermal restriction on I. Put $h = \log\rho'$ and $g(y) = \rho'(\rho^{-1}(y))$. Its energy and variance are*

$$\begin{aligned} E_\rho &= \frac{c}{48\pi}\int h'(x)^2\,dx, \\ \mathrm{Var}_\rho(H_f) &= \frac{c}{48\pi^2}\int_0^\infty \omega^3 |\widehat{g-1}(\omega)|^2\,d\omega < \infty, \end{aligned} \tag{95}$$

where $\widehat{k}(\omega) = \int e^{-i\omega y}k(y)dy$. The global state is pure, not globally KMS. Its outside correlations purify the thermal restriction.

Proof. Set $\alpha = 2\pi T$ and prescribe $\rho = e^{\alpha x}$ on a neighborhood of I. Extend $\log \rho'$ smoothly with compact support. A disjoint bump can adjust $\int (e^h - 1)$ to zero: a positive bump has unbounded positive integral, while a negative bump on sufficiently large support has negative range exceeding any fixed surplus. Then $\rho(x) = x + \int_{-\infty}^{x} (e^h - 1)$ is a compact completion, up to an irrelevant translation on I. Its covariance on I is $iT/[2 \sinh \pi T(x - y - i0)]$. Wick's rule identifies all local fermion correlations; the cited geometric construction gives the corresponding observable-net restriction. Integrating the Schwarzian gives the first moment in Eq. (95). Conjugating H_f into the vacuum gives $T(g) + E_\rho$. Since $H_f \Omega = 0$, the variance is the vacuum second moment of $T(g - 1)$, with the normalization of [11, Eq. (4.11)]. The test $g - 1$ is smooth and compact, so that moment is finite. □

Proposition 44 (Preparation cost in the vacuum orbit)**.** *Every compact-diffeomorphism vacuum preparation with this thermal restriction obeys*

$$E_\rho \geq \frac{\pi c T^2 L}{12} - \frac{cT}{6} \tanh \frac{\pi T L}{2}. \tag{96}$$

The right side is the infimum in this class when outside support is unrestricted, and is not attained for $T, L > 0$.

Proof. Thermal stress gives $\{\rho, x\} = -(2\pi T)^2/2$ on I. Thus $w = h'$ satisfies $w' = (w^2 - \alpha^2)/2$. The branches $w = \pm\alpha$ and the nonsingular coth branches have $w^2 \geq \alpha^2$. The remaining branches are $-\alpha \tanh[\alpha(x - s)/2]$; their derivative-square integral on I is minimized when s is its midpoint. That integral is $\alpha^2 L - 4\alpha \tanh(\alpha L/4)$. For its compact completions, flatten the endpoint derivatives in collars of width ϵ, at cost $O(\epsilon)$, and join the endpoint h-values to zero over length D, at cost $O(D^{-1})$. A disjoint bump of scale D corrects the displacement integral with bounded amplitude and cost $O(D^{-1})$. Taking $\epsilon \to 0$ and then $D \to \infty$ reaches the infimum. Zero exterior cost would force $h' = 0$ outside I, incompatible with its endpoint derivatives. □

This is a preparation-class bound, not a minimum energy for all states or complete devices. Its asymptotic slope is the thermal current. Local negative stress outside I is included; counting only the positive plateau gives a different quantity. For $L = 8$ and $T = 1/(2\pi), .31894305$, explicit smooth completions cost $.11257242, .60868371$, above the respective infima $.02748002, .15992956$. Increasing support or sharpening a boundary can approach the infimum without keeping the drive or rms resources uniformly bounded.

For a regular increasing clock $\eta(v)$ on a finite interval, complete $\rho(v) = \exp[2\pi T \eta(v)]$ instead. The entire local covariance is the thermal η covariance pulled back by that clock, and

$$\langle T(v) \rangle_\rho = \eta'(v)^2 \frac{\pi c T^2}{12} - \frac{c}{24\pi} \{\eta, v\}. \tag{97}$$

Two independently prepared copies therefore supply the two-port input correlations on the operating interval, including the output cross covariance and switching stress. Their global purity and inside–outside correlations remain in the ledger. Thermal local correlations alone do not establish a finite-window erasure bound. The cost in Eq. (95) uses the actual physical coordinate; Eq. (96) cannot be applied to a rapidity interval without its clock conversion. The local diffeomorphism has a driven stress-unitary preparation path. Its autonomous bounded approximation is developed in Appendix I.1.

G.2 A record-selected source is not its average covariance

Suppose a retained classical program R has probabilities p_r, is jointly independent of the two prepared incoming streams, and selects θ_r. Conditional on r, the preceding contact law holds. For a finite frequency regulator with independent input mode pairs, write $\rho_{r,N} = \bigotimes_{j=1}^{N} \rho_{r,j}$ for the reduced source and $\bar{\rho}_N = \sum_r p_r \rho_{r,N}$ for its unconditional state. The return stream and program are retained in the joint process.

Proposition 45 (The source carries the program's relations)**.** *For distinct frequencies j, k, with $\Delta n_j = n_o(\omega_j) - n_H(\omega_j)$,*

$$\mathrm{Cov}(n_j, n_k) = \mathrm{Var}_p(\chi_r)\, \Delta n_j \Delta n_k. \tag{98}$$

Furthermore,

$$S(\bar{\rho}_N) = \sum_r p_r S(\rho_{r,N}) + \mathcal{I}_N, \qquad 0 \le \mathcal{I}_N \le H(R). \tag{99}$$

Here $\mathcal{I}_N = I(R : \text{source})$ in the classical–quantum record state. The product state with the same one-mode occupations has entropy $\sum_j h(\sum_r p_r n_{r,j})$, generally larger. If $H(R) = o(\mathcal{T})$ on a growing observation window and the conditional stationary source entropy rates exist, the unconditional excess rate is $\bar{s} = \sum_r p_r s_{\chi_r}$. Its mean current obeys

$$\bar{q} \ge T_H \bar{s} + \frac{3\bar{s}^2}{\pi}.$$

This limit assumes the same program is retained across the window, not a fresh independently randomized choice at each mode or time.

Proof. Conditional occupations are independent and have means $n_{r,j} = n_H(\omega_j) + \chi_r \Delta n_j$. Conditional covariance therefore vanishes; covariance of their means gives Eq. (98). The entropy identity is the classical–quantum mutual information; its upper bound is the classical program entropy. Subadditivity gives the product-marginal upper bound. Divide the identity by the window length. Averaging Eq. (93) and using $\sum_r p_r s_r^2 \ge \bar{s}^2$ proves the current bound. □

A single persistent program bit can thus create correlations across an arbitrarily long source without contributing an extensive new entropy of its own. Replacing that state by an independent Gaussian state at its averaged covariance destroys those relations. If a new program is acquired at every interval, its entire retained sequence belongs in R; $H(R)$ can then be extensive. Angle switches have the separate two-port work of Eq. (94), and neither entropy identity makes those switches free. These distinctions apply to a gravitational source selected by internal records: mean stress alone does not specify the source state.

There is an exact finite autonomous realization of the static record-controlled exchange. For matched frequencies let $K_j = c_{H,j}^\dagger c_{o,j} + c_{o,j}^\dagger c_{H,j}$ and use

$$V = \sum_r |r\rangle\langle r| \otimes \prod_j e^{-i\theta_r K_j}, \qquad H_0 = \sum_j \omega_j (n_{H,j} + n_{o,j}).$$

It commutes with H_0. Put V on one edge of the positive history clock and identities on the remaining edges. The full Hamiltonian $H_0 + H_{\mathrm{hist}}$ is nonnegative, time independent

and conserves H_0. Its endpoint evolution executes the exchange uniformly over all retained references. This is a finite-mode control realization; its clock initialization and finite thermal preparations are inputs. It does not implement the moving conformal contact's material stress or remove its continuum switching work.

Proposition 46 (Two reservoirs and the complete controller ledger)**.** *Let finite reservoirs start at Gibbs states τ_H, τ_o with inverse temperatures $\beta_H > \beta_o$, jointly independent of a complete zero-bare-gap controller M. Its internal correlations and retained references are included in M, but not purifiers of the reservoirs: those would violate the stated initial independence. Under any joint unitary conserving reservoir bare energy, put $Q = \Delta E_H = -\Delta E_o$, $D_i = D(\rho_i' \| \tau_i)$ and $I' = S(\rho_H') + S(\rho_o') + S(\rho_M') - S(\rho_{HoM}')$. Then*

$$(\beta_H - \beta_o)Q = D_H + D_o + I' - \Delta S_M. \tag{100}$$

If the complete controller entropy is unchanged, the right side is $D_H + D_o + I' \geq 0$. Returning its marginal does not mean eliminating its output correlations.

Proof. The Gibbs identity gives $D_i = \beta_i \Delta E_i - \Delta S_i$. Initial independence and unitary invariance of total entropy give $\Delta S_H + \Delta S_o + \Delta S_M = I'$. Add the Gibbs identities and use energy conservation. This is the multi-reservoir form of the relative-entropy ledger underlying the sharpened Landauer principle [16, Theorem 3]. □

At frequencies $1, 2$, temperatures $.7, 1.4$, program probability .37 and angles $.2, 1.1$, direct joint evolution transfers .13174972 energy into the colder source. Its cross-frequency covariance is .00249817. Source entropy is .86881190, conditional entropy .84854079, and the independent state at averaged occupations has entropy .86900230. A five-state clock has mean energy .12 and interaction cutoff .24; its endpoint matrix and the thermal ledger agree with direct evolution. Both reservoirs have total bare cutoff 6 in these units. None of these cutoffs describes an infinite KMS field.

An operational error also needs an energy qualification before it certifies a source. For $0 \leq H_E \leq E_{\mathrm{cut}} I$, $|\operatorname{tr} H_E(\rho - \sigma)| \leq E_{\mathrm{cut}} D(\rho, \sigma)$. Without a common energy constraint, states at trace distance ϵ can differ by fixed or unbounded energy: mix the vacuum with probability ϵ in occupation $N = 1/\epsilon$. A joint clock or gate-distance certificate alone therefore does not certify stress of an unbounded field. The continuum source and its support require energy-weighted control as well as retained records.

H A physical contact: reaction, source and recoil

A prescribed scattering angle charges its work but leaves its actuator outside the model. Give the angle θ a momentum and inertia $I > 0$, and couple it to a second coordinate y with inertia $J > 0$ and potential

$$V(\theta, y) = \frac{k}{2}(\theta - y)^2 + \frac{K}{2}y^2, \qquad k, K > 0. \tag{101}$$

The coordinates are noncompact; the potential pins them. In the σ_x basis the two-port contact has charges $q_1 = 1, q_2 = -1$. For a smooth, nonnegative profile f_a of integral one,

use the current Hamiltonian

$$H = \frac{p_\theta^2}{2I} + \frac{p_y^2}{2J} + V(\theta, y) + \pi \sum_{i=1}^{2} \int : \left(j_i(x) + \frac{q_i \theta f_a(x)}{2\pi} \right)^2 : \, dx. \tag{102}$$

The currents obey $[j_i(x), j_l(x')] = -i\delta_{il}\delta'(x - x')/(2\pi)$. At oscillator cutoffs, normal ordering uses the positive completed-square Hamiltonian. Its θ^2 counterterm must not be dropped. The continuum identities below apply on a domain where the current and stress products and their regulated limits exist. The positive form constructions below supply both a neutral-current Hamiltonian and the contact on the full fermion carrier. A static current displacement can still have infinite particle number; its dressing and asymptotic state need separate representation control. An oscillator cutoff is not a certificate for the full two-temperature fermion channel. The anomaly-induced reaction is the mechanism of the fermion–rotor model [13, Sections 2.1–2.2]. Here it is applied to the pinned two-coordinate actuator and its stored energy.

Proposition 47 (Current reaction with finite contact width)**.** *Set* $J_i = j_i + q_i\theta f_a/(2\pi)$. *Hamiltonian* (102) *gives*

$$(\partial_t + \partial_x) J_i = \frac{q_i}{2\pi} \dot{\theta} f_a, \qquad I\ddot{\theta} + V_\theta = -\int f_a \sum_i q_i J_i \, dx, \qquad J\ddot{y} + V_y = 0. \tag{103}$$

With the free initial currents defined at $t = 0$*, elimination of the field gives*

$$I\ddot{\theta} + V_\theta + \int_0^t \Gamma_a(t - s)\dot{\theta}(s) \, ds = -\xi_a(t) - \Gamma_a(t)\theta(0), \tag{104}$$

$$\Gamma_a(s) = \frac{1}{\pi} \int f_a(x) f_a(x - s) \, dx,$$

$$\xi_a(t) = \int f_a(x) \sum_i q_i j_i(0, x - t) \, dx.$$

The integrated retarded coefficient is $\gamma = 1/(2\pi)$*. For a Gaussian profile of standard deviation* a*,*

$$\Gamma_a(s) = \frac{\gamma}{\sqrt{\pi} a} e^{-s^2/(4a^2)}, \qquad \operatorname{Re} \widetilde{\Gamma}_a(\Omega) = \gamma e^{-a^2\Omega^2}. \tag{105}$$

Proof. The current commutator gives the first equation in (103); canonical commutators give the other two. Solve the transport equation along its characteristics. The initial $q_i\theta(0)f_a/(2\pi)$ term gives the initial slip in (104), and the source integral gives its convolution. The autocorrelation of f_a is even and has integral one on the whole line, hence $\int_0^\infty \Gamma_a(s) \, ds = 1/(2\pi)$. Fourier transformation gives (105). □

Thus damping is derived from the current algebra, not appended to an externally specified pump. For a velocity history extended into the past with bounded derivative,

$$|(\Gamma_a * v)(t) - \gamma v(t)| \le \|v'\|_\infty \int_0^\infty s\Gamma_a(s) \, ds. \tag{106}$$

The last integral is $2\gamma a/\sqrt{\pi}$ for the Gaussian. If f_a is supported in $[-a,a]$, it is at most $2\gamma a$. This controls smooth histories, not the ultraviolet error of an entire quantum channel. The initial slip is also physical: setting the mean angle to zero does not set its operator fluctuations to zero.

For independent zero-charge thermal inputs with Fermi occupations n_H, n_o, Wick contraction in the original port basis gives the unsymmetrized incoming force spectrum

$$S_\xi(\Omega) = \frac{|\widehat{f_a}(\Omega)|^2}{2\pi} \int_{\mathbb{R}} [n_H(E)(1-n_o(E+\Omega)) + n_o(E)(1-n_H(E+\Omega))]\, dE. \tag{107}$$

Changing variables in $S_\xi(-\Omega)$ yields

$$S_\xi(\Omega) - S_\xi(-\Omega) = 2\gamma\Omega|\widehat{f_a}(\Omega)|^2. \tag{108}$$

Indeed each occupation integrates $n(E) - n(E+\Omega)$ to Ω over the filled sea. The retarded response is temperature independent; its noise is not. Only at equal temperature T does the spectrum reduce to $\Omega|\widehat{f_a}|^2/[\pi(1-e^{-\Omega/T})]$ and obey the single KMS fluctuation–dissipation relation. Different-temperature inputs are generally correlated in the charge basis and cannot be replaced by two independent thermal boson baths.

In a point-contact limit with well-defined renormalized products and the symmetric contact prescription, the total field power and mechanical energy satisfy

$$P_{\text{field}} = \tfrac{1}{2}\{\dot\theta, \xi\} + \gamma\dot\theta^2, \qquad \dot E_{\text{mech}} = -P_{\text{field}}. \tag{109}$$

This follows by substituting the incoming–outgoing current jump into the Sugawara stress and using the canonical mechanical equation. At finite width the full Hamiltonian supplies the energy identity, including retained current energy. The positive prescribed-angle pump term alone is not the power of a quantum actuator. Its velocity correlates with the incoming current. In a stationary interacting Gibbs state $\langle\dot\theta^2\rangle$ can be positive while the net energy transfer is zero.

Proposition 48 (A mechanical record changes the actual field source)**.** *Take a common initial state with zero first moments of the two mechanical coordinates, their momenta and the currents, and finite stress expectations as above. In branch R, displace y by A_R and leave the other first moments unchanged. Relative to the undisplaced evolution under the same Hamiltonian, let $v_R(t)$ be the deterministic mean angle velocity and define*

$$h_R(t,x) = \int_0^t v_R(s) f_a(x-t+s)\, ds, \qquad \Delta\langle T_{\text{field}}(t,x)\rangle = \gamma h_R(t,x)^2. \tag{110}$$

Then

$$\Delta E_{\text{mech}}(t) + \gamma \int h_R(t,x)^2\, dx = \frac{k+K}{2} A_R^2. \tag{111}$$

The actual excess stress scales as A_R^2 for fixed hardware.

Proof. The Heisenberg equations are linear. Displacing the initial y adds the classical solution to the undisplaced operator solution; it leaves every centered correlation unchanged.

Characteristics give $\Delta\langle J_i\rangle = q_i h_R/(2\pi)$. The zero first moments remove the cross term in the stress difference, giving (110). Differentiating its spatial integral gives $v_R(\Gamma_a * v_R)$, the negative derivative of the mechanical displacement energy. The initial displacement cost is $(k+K)A_R^2/2$. □

This cancellation compares displaced and undisplaced evolutions; it does not discard the quantum noise term in (109). The common baseline stress still has to be counted. Energy retained inside the contact profile is not yet energy that has reached an external outlet. An acquired record can select A_R through a paid preparation. That preparation, its storage and its physical copying are additional operations, not free changes of a target potential.

For $I = 10, J = 20, k = 1, K = .1, a = .2$, a 192-mode positive oscillator regulator initially displaced by $A_R = 1.3$ has excess energy .9295. At $t = 80$, .5103950 is in its shifted field coordinates and .4191050 remains mechanical. Direct time-domain integration of the continuum memory equation and separate spatial integration give .5103953 and .4191052, with total energy residual 4.8×10^{-7} at step .005. This is a check of the continuum energy identity, not a cutoff channel certificate. The Gaussian profile has spatial tails. In the regulator's displaced interacting Gibbs state at $T = .02$, $t = 8$, the two branches $A_R = .5, 1.3$ give $\langle \sin^2\theta\rangle = .426644, .716636$. These are coordinate observables, not transmission probabilities for the two-temperature fermion problem. A finite regulated bath eventually recurs.

H.1 Installing a program on the running carrier

Proposition 49 (Autonomous installation on an always-running carrier)**.** *Let $0 \le H_{\mathrm{phys}} \le E_{\mathrm{cut}} I$ be a finite actuator–field regulator, and let a prefix of r gates prepare its mechanical state from an already acquired record. Use a positive history clock with $m \ge r$, strength J, transfer endpoint $T = \pi/(2J)$ and identity edges after the prefix. The common, always-on Hamiltonian*

$$H = H_{\mathrm{hist}} + I_{\mathrm{clock}} \otimes H_{\mathrm{phys}} \ge 0 \tag{112}$$

has, for $t_0 \le t \le T$, joint distance from the completed-prefix state freely continued under H_{phys} for $t - t_0$ at most

$$d_{\mathrm{joint}} \le \min\{1, E_{\mathrm{cut}} t_0 + 2\sqrt{P_0} + 4A_r(t-t_0)\sqrt{P_0}\}, \tag{113}$$

where $P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \le r\}$ and $A_r = J\sum_{j=1}^{r}\sqrt{j(m-j+1)}$. All prior references are retained. For each finite continuation window and specified error, finite sufficiently large clock resources attain that error.

Proof. During the prefix, Duhamel comparison with H_{hist} gives $E_{\mathrm{cut}} t_0$. Completing the prefix on its clock tail costs $2\sqrt{P_0}$. Thereafter compare with the identity-edge clock and the same H_{phys}. Their reference state factors across clock and data, so the nonidentity edges have norm at most $4A_r\sqrt{P_0}$; Duhamel gives the last term. The binomial lower tail decreases up to T. Set $\sin^2 Jt_0 = \mu/m$ with μ a sufficiently large multiple of $\log m$. Then $t_0 \to 0$ and the tail beats the polynomial growth of A_r. Choose T to contain the specified continuation window. □

For a bounded physical source observable $0 \le B \le B_{\mathrm{cut}} I$, the expectation error is at most $B_{\mathrm{cut}} d_{\mathrm{joint}}$. Thus a finite-cutoff record can install a mechanical program while the same carrier and field continue to evolve; no external switch at t_0 is required. A three-level actuator and three-level field, with the old record reference retained, give joint distance .00204248 below the bound .04082928 in one small-clock check. The clock mean energy is 960, so that check is not an installation inside the mass-10 gravitational body. Larger clocks improve the guarantee at larger energy and dimension. The continuum displacement, infrared sector, spatial transport and matching to a moving action need separate control.

The upper energy cutoff is unnecessary for a class of permitted gates. The relevant condition concerns how a gate changes energy, rather than the spectrum of the carrier itself.

Proposition 50 (Energy-regular return on an unbounded carrier)**.** *Let $H_p \ge 0$ be self-adjoint, and let the finite prefix unitaries $W_j = V_j \cdots V_1$, $W_0 = I$, preserve* $\operatorname{Dom} H_p$*. Suppose*

$$B_j = W_j^* H_p W_j - H_p, \qquad b = \max_{j \le r} \|B_j\|, \qquad c = \max_{j \le r} \|[H_p, B_j]\|$$

extend to bounded operators, with B_j preserving $\operatorname{Dom} H_p$*. Let ψ be the normalized full input purification, including prior references, with $E = \|H_p \psi\| < \infty$. Use the same positive history clock as in Proposition 49. For $t_0 \le t \le T$, the joint vectors under $H_{\mathrm{hist}} + I \otimes H_p$ and the reference completed-prefix evolution differ by at most*

$$\varepsilon_m(t) = (E + b)t_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}. \tag{114}$$

Their physical rms energies obey

$$\|H_p \Psi_a(t)\| \le M_a = E + tc + b, \qquad \|H_p \Psi_i(t)\| \le M_i = E + b. \tag{115}$$

These bounds do not grow with the padded clock. If a self-adjoint source observable Q has $\operatorname{Dom} H_p \subseteq \operatorname{Dom} Q$ *and $\|Q\zeta\| \le a\|H_p\zeta\| + d\|\zeta\|$, then*

$$|\langle Q \rangle_a - \langle Q \rangle_i| \le \varepsilon_m(t)\,[a(M_a + M_i) + 2d]. \tag{116}$$

In particular, this certifies the unbounded physical energy $Q = H_p$. For any fixed finite continuation window, $\varepsilon_m \to 0$ with sufficient clock padding.

Proof. The ideal history branches are orthogonal in the clock, and $\|H_p W_j \psi\| = \|(H_p + B_j)\psi\| \le E + b$. The early Duhamel comparison therefore costs $(E + b)t_0$; the prefix tail and late comparison are as in Proposition 49. This gives a vector bound on the retained purification, not only a distance between reduced states.

Put $G = \sum_j |j\rangle\langle j| \otimes W_{\min(j,r)}$ and write the ungated clock as K_m. In the history gauge the total Hamiltonian is

$$G^* H_{\mathrm{tot}} G = K_m \otimes I + I \otimes H_p + \sum_j |j\rangle\langle j| \otimes B_{\min(j,r)}.$$

The last term and its commutator with H_p are bounded by b, c. The propagator commutator consequently has norm at most tc, so the gauge-state rms energy is at most $E + tc$. Transforming back adds at most b. The same domain-preserving commutator argument applies to the reference. Finally write the expectation difference as $\langle \Psi_a - \Psi_i, Q\Psi_a \rangle + \langle Q\Psi_i, \Psi_a - \Psi_i \rangle$ and apply Cauchy–Schwarz. The large-clock limit follows from the same binomial tail choice. □

A positive quadratic-form source needs a weaker estimate. If $0 \le Q \le aH_p + dI$ as forms, Cauchy–Schwarz in the form norm gives

$$|\Delta\langle Q\rangle| \le (\sqrt{aM_a + d} + \sqrt{aM_i + d})\sqrt{a\varepsilon_m(M_a + M_i) + d\varepsilon_m^2}. \tag{117}$$

Here $\|H_p^{1/2}(\Psi_a - \Psi_i)\|^2 \le \|\Psi_a - \Psi_i\|\,\|H_p(\Psi_a - \Psi_i)\|$ proves the bound. A normally ordered local stress is not automatically a positive form or H_p-bounded; its relative estimate must be supplied.

Proposition 51 (A smooth chiral stress certificate)**.** *Consider a positive-energy chiral CFT satisfying the Fewster–Hollands hypotheses, with $H_f = \int T(x)\,dx$ and central charge c_{ch}. Let $0 \le f \in C_c^\infty$ and $\Lambda_f > \|f\|_\infty$. Define*

$$\begin{aligned} q_f &= \frac{c_{\mathrm{ch}}}{12\pi}\int |(\sqrt{f})'|^2\,dx, \\ q_{\mathrm{comp}} &= \frac{c_{\mathrm{ch}}}{48\pi}\int \frac{f'(x)^2}{\Lambda_f - f(x)}\,dx, \qquad d_f = q_f + q_{\mathrm{comp}}. \end{aligned} \tag{118}$$

Then, as quadratic forms,

$$0 \le T(f) + q_f I \le \Lambda_f H_f + d_f I. \tag{119}$$

If $H_f \le a_0 H_p + b_0 I$ as forms for the physical Hamiltonian of Proposition 50, its expectation certificate (117) *applies to the actual local stress $T(f)$ with $a = \Lambda_f a_0$ and $d = d_f + \Lambda_f b_0$. The constant shift cancels between the two states.*

Proof. The chiral quantum energy inequality gives $T(f) \ge -q_f I$ [12, Theorem 4.1]. To obtain the upper bound, one cannot apply that theorem directly to the non-Schwartz function $\Lambda_f - f$. Choose $k \in C_c^\infty$, $0 \le k \le 1$, equal to one on $[-1, 1]$, and set $\chi_L(x) = k(x/L)^2$. For L enclosing $\operatorname{supp} f$, the admissible test $g_L = \chi_L(\Lambda_f - f)$ has QEI constant

$$q(g_L) = q_{\mathrm{comp}} + \frac{c_{\mathrm{ch}}\Lambda_f}{12\pi L}\int |k'|^2.$$

Its two derivative supports are disjoint. On the common CFT core, $T(\chi_L)$ tends in expectation to H_f, using the integrable stress and the translation charge. Thus $T(g_L) = \Lambda_f T(\chi_L) - T(f) \ge -q(g_L)I$ gives the upper bound. The positive form then extends continuously to $\operatorname{Dom} H_f^{1/2}$. Combining it with the stated physical form inequality and (117) proves the certificate. □

For the inertial contact there is a useful conditional transfer of this field bound to the interacting carrier. In its positive completed-square regulator, write the shifted field energy as $\sum_l \omega_l (a_l + s_l\theta)^*(a_l + s_l\theta)$ and put $d_a = \sum_l \omega_l |s_l|^2 = k_{\mathrm{cur}} \int f_a^2/(4\pi)$, where $k_{\mathrm{cur}} = 2$. Since $V \ge kK\theta^2/[2(k + K)]$, the elementary square inequality gives

$$H_f \le 2H_{\mathrm{shifted}} + 2d_a\theta^2 \le \left[2 + \frac{4d_a(k + K)}{kK}\right] H_p. \tag{120}$$

In the neutral current Fock representation, that positive Hamiltonian can be constructed without a static dressing unitary. Set $g_i(\omega) = q_i\sqrt{\omega}\,\widehat{f_a}(\omega)/(2\pi)$, so $\sum_i \int |g_i|^2/\omega = d_a$. On the form domain of $H_0 = H_{\mathrm{mech}} + H_f$, define

$$\mathfrak{q}_H[\psi] = \mathfrak{q}_{\mathrm{mech}}[\psi] + \sum_i \int_0^\infty \left\| \sqrt{\omega}\, a_i(\omega)\psi + \frac{g_i(\omega)}{\sqrt{\omega}}\theta\psi \right\|^2 d\omega. \tag{121}$$

This includes the counterterm. Put $A_a = 2+4d_a(k+K)/(kK)$. The same square inequalities give $\mathfrak{q}_0 \le (1+A_a)\mathfrak{q}_H$ and $\mathfrak{q}_H \le A_a\mathfrak{q}_0$. Thus the two form norms, with $\|\psi\|^2$ added, are equivalent. The closed positive form defines a self-adjoint Hamiltonian, and (120) holds for it. Coupling cutoffs converging in $L^2(d\omega/\omega)$ converge in the common form norm and hence in norm resolvent on this field space. This does not require $g_i/\omega \in L^2$: when $\widehat{f_a}(0) \neq 0$, the static dressing can fail to be a Fock unitary even though the Hamiltonian exists.

For a compact contact profile, a stress test outside its support reads the physical outgoing current stress itself. A Gaussian profile has tails and is not covered by that exact exterior assertion. The gates must be energy-regular for this interacting H_p; free-field regularity alone does not establish that condition when the gate overlaps the contact. The exterior case admits an exact transfer below. This neutral-current construction does not establish a dressed ground state or asymptotic charged representation. The full fermion carrier and its prepared thermal inputs can instead be treated directly. A uniform zero-width limit does not follow from the form inequality.

The contact on the full fermion carrier. The two complex fermions need not be replaced by two neutral current Fock spaces. Work in their full Dirac-sea CAR representation, with positive translation generator H_f. Smooth local gauge implementers and the current anomaly act on this whole space, including its charged sectors [45, Sections 3.2–3.3 and Appendices A–B]. In the σ_x basis write $J(f) = \sum_i q_i j_i(f)$, $\kappa_c = \sum_i q_i^2 = 2$ and $d_f = \kappa_c\|f\|_2^2/(4\pi)$.

Proposition 52 (A positive contact on the full CAR carrier)**.** *For real $f \in C_c^\infty(\mathbb{R})$ and the pinned mechanical potential* (101)*, the form*

$$\mathfrak{q}_p = \mathfrak{q}_{\mathrm{mech}} + \mathfrak{q}_f + \langle \theta J(f)\rangle + d_f\langle\theta^2\rangle$$

is closed and nonnegative on the form domain of $H_0 = H_{\mathrm{mech}} + H_f$. It defines the single-contact self-adjoint H_p on the full fermion space. Put $a_0 = kK/[2(k+K)]$ and $L = \max\{2, 1+2d_f/a_0\}$. Then $L^{-1}\mathfrak{q}_0 \le \mathfrak{q}_p \le L\mathfrak{q}_0$. The assertion does not require a static dressing unitary.

For $b \in C_c^\infty((1,2))$ of integral one, set $b_D(x) = D^{-1}b(x/D)$ and $f_D = f - (\int f)b_D$, with disjoint supports. The corresponding compact-gauge contacts converge in norm resolvent to H_p. If $d_b = (\int f)^2\kappa_c\|b\|_2^2/(4\pi D)$ and $r_D = \sqrt{d_b/a_0} + d_b/a_0$, then, for $Lr_D < 1$,

$$\|(H_{p,D}+1)^{-1} - (H_p+1)^{-1}\| \le \frac{Lr_D}{1-Lr_D}.$$

Proof. First let $f = F'$ with $F \in C_c^\infty$. Multiplication by $e^{\pm i\lambda F}$ has a Hilbert–Schmidt commutator with the one-particle Hardy projection: its squared norm is $\iint |g(x)-g(y)|^2/(x-$

$y)^2\,dxdy/(4\pi^2)$, finite by smoothness and compact support of $g-1$. Thus $W_\lambda = e^{-i\lambda J(F)}$ is implemented on the full CAR space. On the common smooth core, $[J(F), H_f] = iJ(F')$ and $[J(F), J(F')] = i\kappa_c \int F'^2/(2\pi)$ give

$$W_\lambda H_f W_\lambda^* = H_f + \lambda J(f) + d_f\lambda^2 \geq 0.$$

For general f, use f_D above. On a finite smooth particle–hole core, $J(b_D) \to 0$ strongly: its vacuum norm is $O(D^{-1})$, and commuting it past each fermion changes the test function by multiplication with b_D, of supremum norm $O(D^{-1})$. Meanwhile $d_{f_D} \to d_f$. Passing to the limit and minimizing over λ gives the form inequality

$$|\langle\psi, J(f)\psi\rangle| \leq 2\sqrt{d_f}\,\|\psi\|\,\|H_f^{1/2}\psi\|.$$

Polarization extends the form to $\operatorname{Dom} H_f^{1/2}$. For each mechanical coordinate the shifted field form $A_f(\theta) = H_f + \theta J(f) + d_f\theta^2$ is nonnegative and obeys $A_f \leq 2H_f + 2d_f\theta^2$ and $H_f \leq 2A_f + 2d_f\theta^2$. Integrate over the mechanical coordinates and use $V \geq a_0\theta^2$. The displayed equivalent form norms establish closedness and positivity.

For the compensator, the same current bound and Cauchy–Schwarz give

$$|\langle\theta J((\int f)b_D)\rangle| \leq \sqrt{d_b/a_0}\,\mathfrak{q}_0.$$

The counterterm difference is $d_b\langle\theta^2\rangle$. Hence

$$|\mathfrak{q}_{p,D} - \mathfrak{q}_p| \leq r_D\mathfrak{q}_0 \leq Lr_D\mathfrak{q}_p.$$

The resolvent identity in the $(H_p+1)^{1/2}$ form scale gives the stated estimate. □

The finite-rms preparations of Proposition 43 now belong to the same fermion carrier as the inertial contact. Their outside purifications remain present. A constant flavor rotation preserves the vacuum but generally gives off-diagonal covariance for unequal temperatures; the rotated currents are not two independent KMS baths. The force law therefore uses the original quasifree input correlations, not a substitute product of bosonic thermal states. Finite form energy does not by itself supply the graph regularity required by the autonomous return theorem. On the stated smooth input core, exterior even local gates have the same free and interacting commutators by CAR locality, and the local-source certificate uses $H_f \leq LH_p$.

The compensator is a proof and approximation device, not a free heat outlet. For $\int f = 1$, $\operatorname{supp} f \subset [-a, a]$ and $0 \leq s \leq T < D - a$, its retarded kernel differs from the single contact's by at most $2d_b$. For every fixed D, however, $\int f_D = 0$ makes the integrated kernel vanish. Its dc response is zero, whereas the single contact's is $1/(2\pi)$. Norm-resolvent convergence is consequently not a uniform infinite-time heat-transport statement. The single-contact Hamiltonian itself only couples in $\operatorname{supp} f$; it does not require a physical compensator extending to infinity.

These estimates certify smooth, finite-window source charges. To use a sharp or unbounded gravitational kernel, one must prove the relevant source support or tail control. An accurate field gate now has a route to an accurate local source; the source state, its correlations and its gravitational realization still have their own inputs.

The gate class is nonempty inside an accessible algebra. Let $A = \int h(s) e^{isH_p} C e^{-isH_p} ds$ for a bounded self-adjoint C and a real $h \in C_c^\infty$. Its first two commutator norms are bounded by $k_1 = \|h'\|_1 \|C\|$ and $k_2 = \|h''\|_1 \|C\|$. For $V = e^{i\lambda A}$, the energy change $B = V^* H_p V - H_p$ satisfies

$$\|B\| \le |\lambda| k_1, \qquad \|[H_p, B]\| \le |\lambda| k_2 + \lambda^2 k_1^2.$$

Both follow by differentiating $B = \int_0^\lambda e^{-isA} i[H_p, A] e^{isA} ds$. Products retain the bounded-commutator property. Consciousness II's bounded source preparation and shared-pulse result supply this smoothed gate class in a selected shared algebra for its stated translation generator and vacuum [15]. The exterior case below transfers it to the interacting current Hamiltonian. Record-controlled versions act in the record algebra tensored with that selected algebra. Thus an acquired record can install a permitted source without truncating the whole physical spectrum. An exact Weyl displacement need not satisfy the bounded energy-change condition; it is not covered by this assertion. Nor does an infinite KMS field have the finite global rms energy required here.

Proposition 53 (An exterior source gate for the interacting contact)**.** *Use the positive current Hamiltonian* (121) *or the full fermion contact of Proposition 52 with compact profile f_a. Let C be bounded, self-adjoint and localized in a bounded current interval I. Assume it and the free translates $\alpha_s^f(C)$ for $s \in \operatorname{supp} h$ belong to the selected accessible algebra, where $h \in C_c^\infty$ is real. Suppose their localization intervals are disjoint from* $\operatorname{supp} f_a$*. For $A = \int h(s) \alpha_s^f(C) ds$ and $V = e^{i\lambda A}$,*

$$[H_p, A] = [H_f, A], \qquad [H_p, [H_p, A]] = [H_f, [H_f, A]].$$

Both commutators are bounded, with the k_1, k_2 estimates above. The gate is therefore energy-regular for H_p, not only for H_f. Its bounded energy change is the same for both Hamiltonians. Record-controlled versions also meet the autonomous return conditions.

Proof. Write the interaction form as $H_p - H_f = H_{\mathrm{mech}} + \theta \sum_i q_i j_i(f_a) + d_a \theta^2$. Disjointly supported current smearings have zero symplectic pairing; their Weyl algebras commute. Thus A, and the first two derivative smearings defining its free commutators, strongly commute with every $j_i(f_a)$ and act trivially on the mechanical coordinates. The displayed equalities first hold as polarized forms on the common form domain. Free translation smoothing gives bounded commutators; these identities extend through the closed form to $\operatorname{Dom} H_p$ and imply its invariance under V. Integrating the first identity along e^{isA} gives $V^* H_p V - H_p = V^* H_f V - H_f$. The second identity gives the required bounded commutator of this energy change. A finite record control is a block sum of these same identities with a zero-gap record register. □

This class contains a genuine source preparation. Choose a non-scalar localized C in a common accessible algebra, with the local vacuum separating, and a narrow nonnegative h of integral one whose translates stay exterior. Consciousness II's bounded-source proposition then gives $V\Omega$ finite, strictly positive free-field energy for $0 < |\lambda| \|A\| < \pi$. On a finite-rms mechanical state tensored with the current vacuum, that energy is also the change of H_p: the gate commutes with the contact interaction. With the usual local stress domain and integrability assumptions, the source perturbation vanishes outside the translated interval,

and a smooth test equal to one there reads its full positive energy. The autonomous return and QEI bounds certify this preparation on the actual carrier while retaining the acquired record and prior references. A non-scalar shared algebra by itself need not contain such an exterior localized observable; the localization and permission assumptions are part of this construction. It supplies a finite source, not a thermal reservoir, a covariant local history clock or a rule that forms its own permissions.

For an explicit return, take $H_p = N$ on an infinite oscillator and retain three degenerate bits: a held source bit, its acquired record, and an output meter. A controlled NOT first acquires the source bit; that record then rotates oscillator levels $0, 1$ by .2 or .9; a final controlled NOT reads level 1 into the output meter. All higher oscillator levels form a reducing sector. This is an exact unbounded model, not a cutoff approximation. With initial source probabilities $.4, .6$ and its old purification retained, a 513-state clock of mean energy 512 gives vector error .118943 below .199530, and source energy .3839746172 against the reference .3839484296. The energy difference is 2.62×10^{-5}. The actual output-record probability is .3839746172. Clock energy decreases by the source gain; total-energy residual is below 4×10^{-12} in this check. A separate two-level block calculation agrees with the full-register evolution to 2.2×10^{-15} for the smaller clocks. Spatial transport, thermal preparation and the moving support remain distinct from this acquisition–preparation–reading realization.

H.2 The contact position can recoil

The contact profile need not follow a trajectory specified from outside. Let Y be its quantum position, $P_Y = -i\partial_Y$, and $M > 0$ its mass. On the full fermion carrier set

$$U_f(a) = e^{-iaH_f}, \qquad J_Y = U_f(Y) J(f) U_f(Y)^*.$$

The mechanical angle and storage coordinate remain internal to the body. This field translation exists on the full CAR space; it is not the static infrared dressing of the current.

Proposition 54 (Finite-mass contact and momentum conservation)**.** *The form*

$$\mathfrak{q}_{\mathrm{mov}} = \frac{\|P_Y \Psi\|^2}{2M} + \mathfrak{q}_{\mathrm{mech}} + \mathfrak{q}_f + \langle \theta J_Y \rangle + d_f \langle \theta^2 \rangle$$

is closed and nonnegative on the form domain of $P_Y^2/(2M) + H_{\mathrm{mech}} + H_f$. Its self-adjoint Hamiltonian H_{mov} strongly commutes with $P_{\mathrm{tot}} = P_Y + H_f$. A finite anchor Z of mass M_0 can be added with $P_Z^2/(2M_0) + k_s(Y - Z)^2/2$, $k_s > 0$; the conserved momentum then includes P_Z.

Proof. For almost every Y, field translation preserves the current form bound and H_f of Proposition 52. Integration over Y gives the same equivalent form norms after the positive position kinetic energy is added. The relative-anchor form is positive and closed on its natural domain as well. Simultaneous body and field translation sends J_Y first to J_{Y-a} and then to J_Y. All remaining terms, including the relative-anchor potential, are invariant. Uniqueness of the closed form realization implies commutation of H_{mov} with this translation group and therefore strong commutation with its generator. □

Thus the entire total-momentum distribution is conserved. These Hamiltonians measure energy above the body and anchor rest energies; those constant costs must be added in an apparatus ledger. The translation law is unchanged. On a common differentiable core the reaction is

$$\dot{P}_Y = -k_s(Y-Z) - \theta\partial_Y J_Y, \qquad \dot{P}_Z = k_s(Y-Z), \qquad \dot{H}_f = \theta\partial_Y J_Y.$$

The field-energy change is balanced by the body and anchor. Its instantaneous sign is not a certificate of exported heat. A positive record-controlled internal spring can change this dynamics without breaking translation symmetry; its preparation and gate regularity remain separate requirements.

The direct-integral unitary $\mathcal{W}(Y) = e^{-iYH_f}$ gives $\mathcal{W}^* P_Y \mathcal{W} = P_Y - H_f$ and $\mathcal{W}^* J_Y \mathcal{W} = J(f)$. With no anchor, the fixed total-momentum fiber is

$$H_M(p) = H_p + \frac{(p-H_f)^2}{2M}. \tag{122}$$

This is the recoil term of a mobile-field canonical transformation, as in the polaron setting [49]; the current form here is the full-CAR contact above. For a finite anchor put $M_t = M + M_0$, $\mu = MM_0/M_t$ and $r = Y - Z$. Its kinetic terms in a fiber become

$$\frac{(p-H_f)^2}{2M_t} + \frac{[p_r - (M_0/M_t)H_f]^2}{2\mu}.$$

Dropping either recoil square changes the model.

For fixed p, Eq. (122) converges in strong resolvent to H_p as $M \to \infty$. Its positive forms decrease to $\mathfrak{q}_p$ on $\mathrm{Dom}\, H_p^{1/2} \cap \mathrm{Dom}\, H_f$. This intersection contains a core for $\mathfrak{q}_p$, by equivalence with the free mechanical/field form. Resolvent variational minimizers have the closed-form lower bound; approximating a target vector by that core gives a recovery sequence with recoil penalty tending to zero. Strict convexity gives norm convergence of the resolvent vectors. This is not an operator-norm smallness statement. For a static orbit with the indicated finite fourth energy moment, Duhamel gives

$$\|(e^{-itH_M(p)} - e^{-itH_p})\psi\| \le \frac{1}{2M}\int_0^t \|(p-H_f)^2 e^{-isH_p}\psi\|\, ds.$$

Finite mean field energy alone does not meet this orbit condition.

For the unbounded harmonic coordinates used here, there is a sharper limit on a prescribed constant-velocity replacement. Take $p = Mv$ and remove the free kinetic energy $Mv^2/2$. The fiber is $H_p - vH_f + H_f^2/(2M)$, not H_p. Put $v_* = a_0/(a_0 + d_f)$. For $v < v_*$, the form of $H_p - vH_f$ is closed and nonnegative. For $v_* < v < 1$, it is unbounded below on the smooth core. To see both directions, complete the current form with coefficient $1 - v$:

$$H_p - vH_f \;\ge\; H_{\mathrm{mech,kin}} + [a_0 + d_f - d_f/(1-v)]\theta^2.$$

For the reverse statement use the compensated compact gauge of Proposition 52. Its coherent state has $\langle J(f)\rangle = 2d_f\lambda$ and $\langle H_f\rangle = (d_f + d_b)\lambda^2$. Minimizing over λ gives mechanical leading coefficient $a_0 + d_f - d_f^2/[(1-v)(d_f+d_b)]$. First choose finite D large enough to make it negative,

then send the mechanical packet center to infinity, with $y = k\theta/(k+K)$. Fixed packet widths and the conditional gauge derivative have bounded kinetic cost independent of the center. The energy tends to minus infinity. At $v = v_*$ the core form is nonnegative, but the previous equivalent form norm is lost. The full finite-mass H_{mov} remains positive: its recoil square prevents treating body kinetic energy as an unlimited constant-velocity source.

Quantum position also changes the locality requirement for the next thermal preparation. A position wavepacket has tails; its classical center lying in an interval does not localize the whole protocol. Choose a smooth position cutoff χ whose support keeps the full contact profile in a bounded interval I. Let the past interval A contain all free translates of I over the operating time, together with the other field couplings in the interaction picture. Replacing its interaction by $\chi\theta J_Y + d_f\chi^2\theta^2$ defines a positive local comparison Hamiltonian H_A. On a common operator-domain trajectory put

$$\delta_{\mathrm{loc}} = \int_0^t \|(H_{\mathrm{mov}} - H_A)e^{-isH_{\mathrm{mov}}}\Psi\|\, ds.$$

The retained-reference vector error is at most δ_{loc}. The thermal comparison in Appendix I.2 with the *actual* record distribution then has trace error at most $2\delta_{\mathrm{loc}} + \sqrt{I_0/2}$: one locality error changes the joint state, the other changes its record marginal. For source second-moment norms M_a, M_l, M_{pl}, M_{pa} of the actual, local, local-product and actual-product outputs, respectively, its source error is at most

$$\delta_{\mathrm{loc}}(M_a + M_l + M_{pl} + M_{pa}) + \sqrt{I_0}(M_l + M_{pl}).$$

All interaction-picture field couplings of the comparison protocol must lie in A; position-tail probability alone does not bound the generator integral. The body has genuine quantum recoil, but its Schrödinger kinematics does not supply a relativistic causal support, a clock or a dilaton stress tensor.

H.3 Physical locality after unfolding

The preceding H_f translates two co-propagating unfolded fields. Writing physical fields as $\xi_R(x) = \psi_1(x)$ and $\xi_L(x) = \psi_2(-x)$ changes the bulk momentum to $H_1 - H_2$. The finite-width flavor current before diagonalization becomes

$$\int f(x)[\xi_R^\dagger(x)\xi_L(-x) + \xi_L^\dagger(-x)\xi_R(x)]\, dx.$$

It is a mirror coupling, not a local bulk density. The CAR give $[J(f), \xi_L(y)] = -f(-y)\xi_R(-y)$, allowing an immediate response between different positions. Point contact avoids that spatial kernel, but a profile $f_a(x) = a^{-1}f(x/a)$ at fixed integral has $d_{f_a} = d_f/a$. The existing energy bounds are not uniform in that limit. Neither folding nor shrinking supplies a finite local material body for free.

A local current alternative uses two flavors in each physical direction, with charges $+1, -1$ after a within-direction flavor rotation. Let $H_R, H_L \geq 0$, $H = H_R + H_L$, $P = H_R - H_L$, and $d_s = \kappa_s\|f\|_2^2/(4\pi)$, $\kappa_R = \kappa_L = 2$. Both currents $J_R(x), J_L(x)$ now act at the same physical x.

Proposition 55 (Bidirectional current and physical recoil)**.** *The form*

$$H_p^{\mathrm{bi}} = H_{\mathrm{mech}} + H + \theta[J_R(f) + J_L(f)] + d\theta^2, \qquad d = d_R + d_L,$$

is closed and nonnegative on the free form domain. Translating its current by e^{-iYP} and adding $P_Y^2/(2M)$ gives a nonnegative Hamiltonian conserving $P_Y + P$. Its fixed-momentum fiber is $H_p^{\mathrm{bi}} + (p-P)^2/(2M)$. For the harmonic internal coordinates above and equal $d_R = d_L = d/2$, $H_p^{\mathrm{bi}} - vP$ has a positive closed form for $|v| < \sqrt{a_0/(a_0+d)}$ and is unbounded below on the smooth core for $\sqrt{a_0/(a_0+d)} < |v| < 1$.

Proof. In physical coordinates the left current has the opposite signs in both the current commutator and its commutator with H_L. For $f = F'$, $G = J_R(F) - J_L(F)$ therefore satisfies

$$e^{-i\theta G} H e^{i\theta G} = H + \theta(J_R(f) + J_L(f)) + d\theta^2.$$

Compact compensation in each direction handles arbitrary compact f, as in Proposition 52. The same equivalent form norms and translation proof then apply, with P replacing H_f. Completing the two current forms separately gives

$$H_p^{\mathrm{bi}} - vP \geq H_{\mathrm{mech,kin}} + \left[a_0 + d - \frac{d_R}{1-v} - \frac{d_L}{1+v}\right]\theta^2.$$

The bracket is $a_0 - dv^2/(1-v^2)$ in the balanced case. For necessity, use independent compact compensated coherent states in the two directions. Their minimized leading coefficient is $a_0 + d - d_R^2/[(1-v)(d_R + d_{b,R})] - d_L^2/[(1+v)(d_L + d_{b,L})]$. It is negative for fixed sufficiently spread compensators above the stated threshold. Harmonic packet centers tending to infinity then give the negative core sequence, with bounded kinetic cost. □

At equality only core nonnegativity is asserted. The finite-mass Hamiltonian remains positive with its recoil retained. This alternative has four complex chiral fields, $c_R = c_L = 2$; each actual outlet must count the fields routed through it. Four input preparations, their purifiers and the additional control permissions are real resources. For interval lengths a, b and separation $g > 0$ in the product free vacuum, Appendix I.2 uses $I_0 = \frac{2}{3}\log[1 + ab/(g(g + a + b))]$ for these four fields, twice the two-field value. Both the conditional preparation error and the source-moment certificate must use this larger I_0. The balanced free carrier and its physical recoil do not yet quantize a propagating material controller or its complete stress tensor.

I Local preparation and correlated-window renewal

I.1 A local preparation can be executed autonomously

The compact preparation unitary V of Proposition 43 is local, but its energy change need not be bounded. A preparation bit permits a bounded regularization that controls the field state in energy norm as well as Hilbert norm. Let H act trivially on that zero-gap bit and set

$$C = |1\rangle\langle 0| \otimes V + |0\rangle\langle 1| \otimes V^*, \qquad u_0 = |0\rangle\Omega, \quad u_1 = Cu_0.$$

Then $C = C^*$, $C^2 = I$ and $e^{-i\pi C/2}u_0 = -iu_1$. For a nonnegative even $h \in C_c^\infty(-1,1)$ of integral one, put $\mu_2 = \int s^2 h(s)ds$ and

$$C_\delta = \int h(s) e^{i\delta s H} C e^{-i\delta s H} ds, \qquad W_\delta = e^{-i\tau C_\delta}, \quad \tau = \pi/2.$$

Proposition 56 (Local preparation in the energy graph norm)**.** *Suppose $F_l(t) = e^{itH}Ce^{-itH}u_l$, $l = 0, 1$, is twice continuously differentiable in the H graph norm for $|t| \le \delta_0$. Let $L_2 = \max_{l,t} \|F_l''(t)\|$ and $G_2 = \max_{l,t} \|HF_l''(t)\|$ there. For $0 < \delta \le \delta_0$ define*

$$e_\delta = \mu_2 \delta^2 L_2/\sqrt{2}, \qquad g_\delta = \mu_2 \delta^2 G_2/\sqrt{2}, \qquad k_1 = \|h'\|_1/\delta, \quad k_2 = \|h''\|_1/\delta^2.$$

The bounded local gate W_δ has energy-regular constants $b \le \tau k_1$ and $c \le \tau k_2 + \tau^2 k_1^2$. Moreover,

$$\begin{aligned} \|(W_\delta - e^{-i\tau C})u_0\| &\le \tau e_\delta, \\ \|H(W_\delta - e^{-i\tau C})u_0\| &\le \tau g_\delta + \tfrac{1}{2}\tau^2 k_1 e_\delta. \end{aligned} \tag{123}$$

For a compact diffeomorphism of the vacuum Virasoro module the stated graph condition holds for free translation $H = H_f$. For every finite source tolerance, a finite positive autonomous history clock can therefore execute this preparation while approximating its local thermal restriction and each specified smooth stress test.

Proof. Time smoothing gives $\|[H, C_\delta]\| \le k_1$ and $\|[H, [H, C_\delta]]\| \le k_2$, hence the gate constants by Proposition 50. Evenness of h cancels the first Taylor term. The second-order remainder bounds $(C_\delta - C)u_l$ by $\mu_2\delta^2 L_2/2$ and its H norm by $\mu_2\delta^2 G_2/2$. The ideal gate path lies in $\operatorname{span}\{u_0, u_1\}$ with coefficient sum at most $\sqrt{2}$. Duhamel gives the first estimate. For $\zeta \in \operatorname{Dom} H$,

$$\|He^{-itC_\delta}\zeta\| \le \|H\zeta\| + tk_1\|\zeta\|.$$

Inserting this in the same Duhamel integral gives the second estimate, which tends to zero as $O(\delta)$. The smooth diffeomorphism representation preserves $\bigcap_n \operatorname{Dom} L_0^n$, and its action is smooth there [12, Sections 5.1–5.2, Proposition 5.1]. Möbius translation generators act continuously on this domain; Ω, $V\Omega$ and the required orbit therefore have the stated graph differentiability. First make the preparation error small by choosing δ, then, for fixed δ, apply the autonomous return and smooth-stress certificates. The required clock is finite at each tolerance, although its resources need not stay bounded as that tolerance decreases. □

No global vacuum or energy projector enters C_δ. Its support is the preparation interval enlarged by the short translations. If this interval stays outside the pinned-current contact, the commutator equalities above make W_δ energy regular for the same interacting H_p. Graph convergence also needs its H_p-graph condition on the input; commutator equality alone does not give it. Schwartz mechanical vectors tensored with the smooth vacuum core supply an explicit class: the mechanical polynomials, smeared currents and twice-differentiated exterior orbit act continuously on that core. Overlapping intervals and more general mechanical states need separate domain control.

In the free-field execution, field-energy gain is exactly the loss of the clock-interaction energy $\langle H_{\text{hist}} \rangle$; their sum is conserved. The preparation bit, initial vacuum, allowed local

gate, clock coherence and waveform are supplied resources. Locality here is locality of a permitted continuum-algebra interaction, not a derived laboratory actuator or a covariant moving clock. A finite record can select among these gates while retaining its coherences and references. Online prediction of its changing thermal clock and physical renewal remain further requirements.

I.2 A retained record can select the next thermal window

A later interval of the same field is not an independent reservoir merely because it has not yet met the contact. The vacuum correlates disjoint regions; a record made in one can steer the conditional state in the other. Separation nevertheless gives a quantitative renewal condition without discarding those correlations.

Proposition 57 (Record-selected local thermal preparation)**.** *Consider c_f independent complex chiral fermions in their vacuum. Let A, B be split-separated intervals of lengths a, b and gap $d > 0$. The complete past field protocol and acquired record R are a normal local channel on A, with initial ancillas independent of the field. Here R denotes the commuting readout algebra of the retained register, not an extra physical dephasing. For each result r, let an even $V_r \in \mathcal{A}(B)$ prepare a specified thermal restriction τ_r on $J \subset B$ from the vacuum. Put*

$$I_0 = \frac{c_f}{6} \log \left(1 + \frac{ab}{d(d+a+b)} \right).$$

After record-controlled preparation,

$$\frac{1}{2} \left\| \rho'_{RJ} - \sum_r p_r |r\rangle\langle r| \otimes \tau_r \right\| \leq \sqrt{I_0/2}. \tag{124}$$

The norm is that of normal states on the local algebra. The bound is a probability-weighted conditional error, not a uniform bound for arbitrarily rare results.

Let Q_r be a smooth source test on J and $Q = \bigoplus_r V_r^* Q_r V_r$. On the split algebra set $\alpha = \rho_{RB}$, $\beta = \rho_R \otimes \omega_B$ and suppose $M_a^2 = \alpha(Q^2)$, $M_i^2 = \beta(Q^2)$ are finite. Then

$$|\alpha(Q) - \beta(Q)| \leq \sqrt{I_0}(M_a + M_i). \tag{125}$$

Neither assertion replaces the actual outside purification by a product state.

Proof. The past local channel leaves the B marginal equal to ω_B. Relative-entropy data processing gives $S(\alpha\|\beta) = I(R : B) \leq I_{\mathrm{vac}}(A : B)$. For free complex chiral fermions, the multi-interval vacuum entropy formula gives the displayed I_0 [46, Section 4.1, Eq. (63)]. For commuting even observable restrictions it supplies an upper bound by data processing; the graded fermion fields are not being replaced by bosonic tensor factors. Measuring an arbitrary binary effect and applying classical Pinsker gives $\|\alpha - \beta\|/2 \leq \sqrt{I_0/2}$. Controlled preparation and restriction are contractions, proving Eq. (124).

For the source bound use the natural-cone representatives ξ_α, ξ_β. Spectral Jensen for the relative modular operator gives [47, Section 2, Eqs. (2.1)–(2.6)]

$$S(\alpha\|\beta) \geq -2 \log \langle \xi_\alpha, \xi_\beta \rangle \geq \|\xi_\alpha - \xi_\beta\|^2.$$

Faithful regularization gives the same bound for finite-entropy normal states. Expanding the two expectations of Q and applying Cauchy–Schwarz gives Eq. (125). □

The estimate can also retain more than the classical record. If the past protocol outputs a larger controller algebra $\mathcal{K}$, the same data-processing argument gives $S(\rho_{\mathcal{K}B}\|\rho_{\mathcal{K}}\otimes\omega_B) \le I_0$. The controller may include its mechanics and references initially independent of the field. Any subsequent normal joint channel on $\mathcal{K}$ and B preserves the local trace bound. Its output source obeys Eq. (125) with the actual and comparison output moments. This also controls a later joint contact, provided its participating algebra is genuinely contained in this input. It does not include the entire field complement as an independent reference.

The source moments are explicit inputs. For the record-only estimate, if the past instrument has local effects $E_r \in \mathcal{A}(A)$ and $m_r^2 = \omega((V_r^* Q_r V_r)^2)$, then locality and $0 \le E_r \le I$ give $M_a^2 \le \sum_r m_r^2$ and $M_i^2 = \sum_r p_r m_r^2$. Smooth compact preparations and stress tests have finite m_r. A trace-distance bound alone would not control an unbounded source.

For a desired local trace error $\epsilon > 0$, a sufficient separation is

$$d \ge \frac{\sqrt{(a+b)^2 + 4ab/(e^{12\epsilon^2/c_f} - 1)} - (a+b)}{2}.$$

This is the gap required by the vacuum-information certificate, not a necessary distance for every instrument: an uninformative record can already be independent at a smaller gap.

For a specified finite sequence, use the actual hull of the past field operations at each step and a separated preparation support for the next window. The estimate applies to that actual past, without resetting it. The local gates can be smoothed and executed by the positive history clock under the graph-domain conditions above; its additional gate and source errors are charged separately. All prior registers and field purifications remain present. The gap consumes spatial reach and transport time. The record must reach the preparer through a permitted physical channel; a controlled unitary does not itself construct that transport. The growing supports, storage and clock therefore define a finite operating family, not a fixed-capacity perpetual reservoir. This local certificate also does not prove that the whole future field and every old quantum reference are jointly a product state. Nor does it alone certify the entropy rate of a correlated released field.

J Actual diagnostic return and terminal reading

J.1 The reading itself supplies the repair and contact program

The program need not be a label given to the apparatus. Use three zero-gap pointer bits for world, sensor and record frame, and three meter bits. Start the routine meter blank and the other two meters in independent diagonal states with bit probability .1. Three controlled NOTs copy body parity into the routine meter; two copy sensor and frame values into their respective noisy meters. All six bits and the actual held body remain in the core.

The acquired meter word o controls a fixed repair table $a(o)$. The table is the first two-step decision of Consciousness III's continuing-body model [14], with independent coordinate fault

probability .05. It depends on the acquired word and known model, not on the inaccessible true cause. On one history edge use the unitary

$$V = \sum_{o} |o\rangle\langle o| \otimes X_{a(o)} \otimes S_{\theta \mathbf{1}_{a(o)\neq 0}}, \tag{126}$$

where $X_0 = I$ and $X_{e_j} = X_j$. The exchange S acts on two matched-frequency fermion cells initially prepared at temperatures $.7, 1.4$, with $\omega = 1$ and $\theta = .7$. Thus the actual physical repair permission and actual energy transfer are performed by one program-controlled history edge. Whether the scorer later calls the repair correct has no effect on the interaction.

Put the five meter gates and this gate on the first six edges of an eight-edge clock, followed by two identities. Every gate commutes with reservoir bare energy. The joint positive conserved Hamiltonian has eight qubit registers, nine clock states and dimension 2304. At $J = .06$, clock mean energy is .48, interaction cutoff .96, and the total spectral cutoff is 2.96. Initial mean energy is 1.00197392. Direct evolution retaining 128 purification columns agrees with the full gate product to vector error 2.7×10^{-14}; bare energy conservation residual is 4.5×10^{-16} in this check. Actual restoration failure is .02891, repair probability .1355, and cold-port energy gain is .00761031. The latter equals $\Pr(a \neq 0)\sin^2\theta(n_o - n_H)$, not a probability of incorrect repair supplied as a source command. The contact angle is a hardware choice: setting it to zero leaves the logical repair intact. This construction connects actual repair to an actual source; it does not make that heat transfer a universal minimum cost of repairing a pointer. The complete six-bit controller entropy is unchanged. Reservoir relative entropies are .0001838782 and .0001317734, and joint output total correlation is .0051202854. They obey Eq. (100) with residual below 3×10^{-16}. Reservoir purification columns are retained for the vector certificate, not folded into its initially independent controller in the entropy ledger.

This is a first-interval conserved realization of acquisition returning to repair and a source. Initial meter purity, biased noise, reservoir preparation, reference labels and hardware are supplied and charged separately from the finite Hamiltonian's energy. The two cells store transferred energy; spatial release and a covariant supported contact are not implied by this finite calculation. Its value is that reading, decision, repair and source no longer need mutually incompatible clocks or an external selector within the demonstrated interval.

Proposition 58 (A finite continuing diagnostic return)**.** *For every specified horizon L, the rational pointer model with $h = 1/20$, $\epsilon = 1/10$ and the two-step decision admits one finite, time-independent, nonnegative Hamiltonian. Its completed circuit has the joint law of all L logical intervals: the same damaged and repaired body, all actual meter and action records, exact retained posterior weights, and one record-selected thermal exchange per interval. No inaccessible fault bit enters the decision circuit. With a fresh matched-frequency cell pair of gap ω per interval,*

$$Q_L = \omega \sin^2\theta(n_o - n_H) \sum_{t=1}^{L} \Pr(a_t \neq 0). \tag{127}$$

For a circuit with m padded history edges, clock mean energy is Jm, the total energy cutoff is $2L\omega + 2Jm$, and its exact completion time is $\pi/(2J)$ in units $\hbar = 1$.

Proof. Prepare $3L$ independent diagonal fault ancillas and let each control the corresponding held-body flip. Prepare $3L$ meter bits as above. The observer's reversible decision uses only the meter words, retained integer weights and known earlier actions. The rational update has denominator 800000 per interval; all weights and comparisons have finite integer size. A reversible Boolean circuit computes each update and action, keeping its inputs and garbage. The action controls the held-body flip and exchange with the next prepared thermal pair. Every gate commutes with the sum H_0 of cell energies, because the logic is degenerate and each exchange has matched gaps. Put the gates on a positive history clock. The prefix-unitary conjugation gives $H_0 + JmI + 2JJ_x$, proving positivity, bare-energy conservation and full-reference endpoint completion. Each unused pair is still jointly independent of the acquired program at its contact. Conditional energy transfer is zero or $\omega \sin^2 \theta(n_o - n_H)$, which proves Eq. (127). □

For the first three logical intervals, the expected number of repairs is .44635607 and $Q_3 = .02506944$ at the stated cell settings. There are nine readings, nine prepared fault bits and six thermal cells. Keeping all preceding weight vectors requires at most 984 bits in this case, before records, arithmetic workspace and reversible garbage. The general saved-weight cost grows quadratically with L. This construction removes the need for a new externally chosen circuit at every diagnostic return. Its logical intervals are not equally spaced proper-time samples of a continuously acting bath: only the completed joint law is asserted. All ancillas and cells are prepared before the finite run, all outputs remain in its ledger, and the clock can reverse after completion. It proves neither a fixed-capacity cyclic apparatus nor a covariant moving contact. These are distinct renewal and material implementation questions, not failures of the finite return.

J.2 An actual terminal record and probe replacement

Let probe P retain any old reference, record R start in $|0\rangle$, and a paid blank B start in $|+\rangle$. Write $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$, then apply the true SWAP_{PB}. The record-one probability is $\mathrm{tr}(P_+^Y \rho_P)$. The final probe is $|+\rangle$, independent of all old references and the record; its old correlations have moved into B and R, not disappeared. Positive generators $gP_+^Y \otimes (I - X_R)$ and $g(I - \mathrm{SWAP}_{PB})$ implement these gates at duration $\pi/(2g)$. A fermionic exchange is not a substitute for this true SWAP.

The physical probe may continue to acquire a phase while the meter executes. Move the r nonidentity gates of the clock in Appendix E to its last r edges. All earlier edges are identities. Put $T = \pi/(2J)$, choose $t_0 < T$ and define

$$P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \geq m - r\}, \qquad A = J \sum_{j=m-r+1}^{m} \sqrt{j(m-j+1)}.$$

Proposition 59 (Terminal read under a retained phase coupling)**.** *Let the actual probe coupling be $\lambda d(t) Z_P$, with $|d(t)| \leq D$. Relative to writing the probe's actual phase at t_0 and replacing it at T, the full joint half trace distance obeys*

$$d_{\mathrm{joint}} \leq 2At_0\sqrt{P_0} + 2\sqrt{P_0} + \lambda D(T - t_0). \tag{128}$$

All old references, the blank, record and clock are included. For any specified finite trajectory and positive error tolerance, finite m and $t_0 < T$ suffice.

Proof. Before t_0, compare with the identity-edge clock and the actual free probe phase. Only the last r edges differ. Their action on that reference is bounded by $2A\sqrt{P_0}$, since the binomial upper tail increases up to T. Duhamel gives the first term. At t_0, dressing the reference with the gate prefixes changes its joint state by at most $2\sqrt{P_0}$. From there to T, omit only the bounded phase coupling, at cost $\lambda D(T-t_0)$; the compiled clock completes the meter and SWAP exactly. First choose t_0 near T, then increase m to suppress the tail. □

The positive clock has mean energy Jm, not zero. Add the probe/blank offset $2\epsilon_P I$, with $2\epsilon_P > \lambda D$, to keep the complete internal Hamiltonian positive. For $m = 10^5$, $r = 2$, $T = 80$, $t_0 = 78$ and $\lambda D = .00032$, a sufficient error is below .000641; clock mean energy is 1963.4954. This cannot be inserted into a mass-10 support without changing that support. A direct 17-state clock check gives joint distance .0034887 below its certificate .0046290. The construction renews the probe over a finite window. It consumes a coherent blank and a record slot and does not reset the entire reader or its clock. Its phase frame and allowed gate family remain inputs. HAFF D also solves a smaller, 129-state meter clock jointly with a source-matched mean moving support [23]. Its energy is included in the initial supported mass, and the actual record depends on the dilaton trajectory. That mean-field calculation does not include the contact reservoir's microscopic degrees or stress. For the specified C_Y, SWAP and coherent blank, an antiunitary symmetry makes the mean probe force vanish throughout the read, although the force operator and its fluctuations remain nonzero.

K Local linear-current control of stress flows

Consider the neutral current on a circle of length 2π, with

$$J_n = \sqrt{n}\, a_n, \quad J_{-n} = \sqrt{n}\, a_n^*, \quad J_0 = 0, \qquad H = L_0 = \sum_{n\geq 1} n a_n^* a_n.$$

Our Fourier convention is $f_n = (2\pi)^{-1}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta$. The same convention applies to the real current smearing h. The primitive interaction is $-Bj(h)$, where $j(h) = \sum_{n\geq 1}\sqrt{n}(h_{-n}a_n + h_n a_n^*)$ and B is a finite controller coefficient. Fix $h \in C_c^\infty(I,\mathbb{R})$ on a proper arc I, with $h_n \neq 0$ for $n \geq 1$. Such an h exists: in the Fréchet space of real smooth functions supported in a fixed compact subarc, each equation $h_n = 0$ defines a proper closed subspace, so the intersection of their open dense complements is nonempty. For a prescribed finite cutoff N, a nonnegative sufficiently narrow bump already gives $h_n \neq 0$ for $1 \leq n \leq N$; this alternative makes the probe depend on N. No uniform lower bound on $|h_n|$ is assumed.

Proposition 60 (A neutral-current stress compiler)**.** *Suppose signed local linear-current coupling, finite degenerate-helper rotations and record phase gates, calibrated clock preparation and finite readout are available, with coherent control intervals. For every real smooth f, $t \in \mathbb{R}$, $E < \infty$ and $\delta > 0$, a finite positive autonomous program implements $e^{itT(f)}$ within*

joint half-trace distance δ, uniformly over inputs with $\operatorname{tr}(H\rho) \le E$ and arbitrary retained references. Its helpers and original clock are included in the comparison with their designated final states. Here

$$T(f) = \frac{1}{2} \sum_{p,q \in \mathbb{Z}} f_{-p-q} : J_p J_q :$$

is the $c = 1$ Sugawara stress tensor. The program uses no field reset, heralding or primitive quadratic coupling.

Proof. We first include the field's free propagation. For a finite quadrature $Q(v) = \sum_{n \le N} (v_n a_n^* + \overline{v_n} a_n)$ put

$$G_n = \frac{v_n}{\sqrt{n} h_n}, \qquad g_v(u) = \frac{1}{2\pi} \sum_{n \le N} (G_n e^{-inu} + \overline{G_n} e^{inu}), \quad 0 \le u \le 2\pi.$$

The Hamiltonian

$$H - s g_v(u) A j(h) + s^2 g_v(u)^2 C_h, \qquad C_h = \sum_{n \ge 1} |h_n|^2,$$

is positive by completion of the oscillator squares, with shifts $s g_v(u) A h_n / \sqrt{n}$ and $A = X$ or Y. In the interaction picture $[j_u(h), j_w(h)]$ is a scalar. Thus the Magnus series terminates after its scalar second term, and $e^{-2\pi i H} = 1$ gives the endpoint gate $e^{i\phi_s} e^{isAQ(v)}$. The phase ϕ_s is computed from the scalar commutator and the completion integral. Subsequent pulses start at integer periods and therefore act on the same $Q(v)$. Changing s to $-s$ pays for the inverse without reversing H. A Y pulse may instead use a paid Z rotation conjugating the X coupling. For coherent record branches, the known phases are removed by record phase gates.

The exact trigonometric comparator does not have smooth compact-time switching. Multiplying it by an all-order-flat endpoint taper χ_η makes the physical coupling smooth and gives $\|(1 - \chi_\eta) g_v\|_1 \le 2\eta \|g_v\|_\infty$. Writing $C_1 = \sum n |h_n|^2$ and $D_h = \sum n^2 |h_n|^2$, we have

$$\|j(h)\psi\| \le 2\sqrt{C_h} \|H^{1/2}\psi\| + \sqrt{C_1} \|\psi\|, \qquad \sqrt{\langle H \rangle_u} \le \sqrt{E} + \sqrt{D_h} |s| \|g_v\|_1.$$

The second bound follows directly from the controlled displacement. It includes the transient excitation of all modes, not just the selected endpoint modes. Duhamel's formula then makes each taper error arbitrarily small in joint vector norm, including old references. For a finite word of such pulses the energy bound is $(\sqrt{E} + \sqrt{D_h} L)^2$, where $L = \sum_{\text{slots}} |s| \|g_v\|_1$.

Next use the current-factor cutoff

$$T_N(f) = \frac{1}{2} \sum_{0 < |p|, |q| \le N} f_{-p-q} : J_p J_q : .$$

This is a selfadjoint finite oscillator quadratic. Finite occupation vectors are analytic for a finite quadratic, hence form an essential selfadjointness core. Let $P_M = 1_{[0,M]}(H)$, M an integer. It has finite rank. For $N > M$, the omitted number-preserving terms have annihilator index $m \le M$ and creator index $n > N$. Their norm on P_M is bounded by

$\sqrt{nM}|f_{n-m}|$. Omitted pair annihilators vanish; pair creators have norm at most $M+2$. Consequently

$$\|[T(f)-T_N(f)]P_M\| \le b_N(M) \longrightarrow 0,$$
$$b_N(M) = \sqrt{M}\sum_{m=1}^{M}\sum_{n>N}\sqrt{n}|f_{n-m}| + \frac{M+2}{4}\sum_{k>N}k^2|f_k|.$$

For the pair term we used $\sum_{n=1}^{k-1}\sqrt{n(k-n)} \le k^2/2$. These absolutely convergent sums identify the limit on the finite-energy core as the Sugawara $T(f)$. Carpi–Weiner's essential selfadjointness theorem for real smooth stress smearings [2, Theorem 4.4] supplies the self-adjoint closure of the limit; it does not substitute for the current-factor tail estimate. On this common core, the resolvent identity and density of $(T(f)-z)D_{\mathrm{fin}}$ give strong resolvent convergence. Hence $e^{itT_N(f)} \to e^{itT(f)}$ strongly, and

$$a_N(M,t) = \|[e^{itT_N(f)} - e^{itT(f)}]P_M\| \longrightarrow 0.$$

The uniform reference estimate is obtained only once, at the input of the whole program. For isometries V, W, purification and the P_M decomposition give

$$D_{\mathrm{joint}}(V\rho V^*, W\rho W^*) \le \|(V-W)P_M\| + 2\sqrt{\frac{E}{M+1}}.$$

It is independent of reference dimension. Thus the factor-cutoff unitaries converge uniformly over the stated mean-energy class. The core-tail bound is not itself a rate for $a_N(M,t)$, since a stress-evolved vector need not stay in P_M.

Write $r = (q_1, p_1, \dots, q_N, p_N)$. A real symmetric matrix K_N gives

$$T_N = \tfrac{1}{2}r^T K_N r - c_N, \quad c_N = \tfrac{f_0}{2}\sum_{n\le N} n, \quad T_N = \tfrac{1}{2}\sum_{\ell\le 2N}\kappa_\ell Q_\ell^2 - c_N$$

after real orthogonal diagonalization. This is an algebraic decomposition, not a free optical transformation: each Q_ℓ is selected directly by the local pulse above. Finite quadratic Trotter products converge on oscillator Schwartz vectors and hence strongly, including after restriction to P_M. Each square exponential is compiled with the same retained helper by four Pauli pulses. For an exponent of magnitude $|\alpha|$, n blocks and $s = \sqrt{|\alpha|/(2n)}$, the second-moment estimate is

$$D_{\mathrm{joint}} \le (13\sqrt{2})^{1/3}\sqrt{|\alpha| m_2}\, n^{-1/6}, \qquad m_2 = \mathrm{tr}(\rho Q_\ell^2) \le 4\,\mathrm{tr}(H\rho) + 1.$$

To see the moment bound, on a spectral value q set $C_\vartheta = e^{i\vartheta X}e^{i\vartheta Y}e^{-i\vartheta X}e^{-i\vartheta Y}$. Taylor's integral remainder gives $\|C_\vartheta - e^{-2i\vartheta^2 Z}\| \le 13|\vartheta|^3$: the third derivative of the four-factor product has norm at most 64, and the quadratic exponential contributes at most $2\vartheta^4$ for $|\vartheta| \le 1$; outside that interval the unitary distance is at most 2. Telescoping n blocks yields a spectral error $\min\{2, A|q|^3\}$, with $A = 13|\alpha|^{3/2}/(2\sqrt{2n})$. The inequality $\min\{4, A^2q^6\} \le 4^{2/3}A^{2/3}q^2$ and the spectral theorem give the displayed joint bound, retaining the helper and reference. The sign is chosen by the Pauli order; all inverse pulses are paid. Moments for

a finite word are evaluated on its ideal prefixes, whose oscillator moments are finite on P_M. Telescoping the complete word against those prefixes avoids both fresh-helper resets and a circular block-count estimate.

Finally approximate the smooth finite schedule in L^1 by constant slots with rational history-clock thresholds. Every slot has the positive field Hamiltonian $H - B_j j(h) + C_h B_j^2$. Install the finite rotations and record phase gates on the original clock path; these gates commute with the degenerate data bare energy. The resulting sum of the positive history clock, data energy and completed field squares is a positive autonomous Hamiltonian on the common field-energy domain. For $G = \max_j \|B_j\|$ and $D_3 = \sum n^3 |h_n|^2$, its field rms obeys

$$D^+ \|H\psi_u\| \le 2G\sqrt{D_h}\sqrt{\|H\psi_u\|} + G\sqrt{D_3}.$$

The bound is independent of clock size. The finite-boundary history-clock estimate therefore makes the complete endpoint error arbitrarily small on P_M, with the same original clock and all record branches retained. Choosing successively M, N, Trotter length, square-block counts, taper/slot precision and clock size allocates any prescribed total δ. □

The resources can grow sharply. With $\Gamma(v) = \|g_v\|_1 \le 2\sum_{n\le N} |v_n|/(\sqrt{n}|h_n|)$, a square exponent of size $|\alpha|$ uses field duration $8\pi n$, area $2\sqrt{2n|\alpha|}\Gamma(v)$ and positive completion integral

$$\frac{2C_h|\alpha|}{\pi} \sum_{n\le N} \frac{|v_n|^2}{n|h_n|^2}.$$

Helper rotations, phase gates and readout have their own finite budgets; rotation slots can last 2π so the field drift is identity. For an autonomous clock of duration T and size $m+1$, $J = \pi/(2T)$, its initial mean energy is mJ, norm $2mJ$ and largest edge coefficient at most $J(m+1)/2$. Preparation, drive bandwidth and retained storage are additional resources. These are sufficient finite bounds, not optimal costs or net-heat identities.

The uniform channel statement assumes only mean energy. Expectations of unbounded stress require separate graph-norm control: for smooth g, $\|T(g)\psi\| \le C_g\|(H+1)\psi\|$. Small joint channel error alone does not certify source convergence on the whole mean-energy class. The finite program's field moments also do not include a covariant stress tensor for clock and support.

The target preserves the Virasoro vacuum subspace, although the individual current pulses and finite T_N approximants can leave it; the endpoint leakage is part of the error above. The compiler concerns the neutral current sector. Its controls are not invariant under the $SO(3)$ action in the $SU(2)_1$ carrier, so an always-on selecting noise needs a separate control/noise certificate. The circle, calibrated port, coherent interval and controller preparation remain physical inputs.

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